

Introduction to Riemann surfaces

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1 Introduction: Elliptic curves

Definition 1.1 *A complex manifold of dimension n is a differential manifold (which we suppose Hausdorff and second countable) equipped with a cover $M = \cup U_\alpha$ by open sets U_α and homeomorphisms (charts) $z_\alpha : U_\alpha \rightarrow \mathbb{C}^n$ such that the maps $z_\beta \circ z_\alpha^{-1} : z_\alpha(U_\alpha \cap U_\beta) \rightarrow z_\beta(U_\alpha \cap U_\beta)$ are biholomorphisms.*

We will usually consider a maximal family of charts compatible with a given chart cover.

Definition 1.2 *A Riemann surface is a one dimensional complex manifold.*

Example 1.3 *The Riemann sphere $\mathbb{CP}^1$ is a Riemann surface whose underlying topological manifold is the two dimensional sphere S^2 . We write $S^2 = \mathbb{C} \cup \{\infty\}$. There are two natural charts:*

1. $z_1 : \mathbb{C} \cup \{\infty\} \setminus \{0\} = U_1 \rightarrow \mathbb{C}$ defined by $z_1(z) = 1/z$ if $z \neq 0$ and $z_1(\infty) = 0$
2. $z_2 : \mathbb{C} = U_2 \rightarrow \mathbb{C}$ defined by $z_2(z) = z$

In the intersection $U_1 \cap U_2 = \mathbb{C} \setminus \{0\} = z_1(\mathbb{C} \setminus \{0\}) = z_2(\mathbb{C} \setminus \{0\})$ we obtain

$$z_2 \circ z_1^{-1} : \mathbb{C} \setminus \{0\} \rightarrow \mathbb{C} \setminus \{0\}$$

given by $z_2 \circ z_1^{-1}(z) = 1/z$ which is a biholomorphism.

This is the most symmetric example. It has the largest group of automorphisms (the group of diffeomorphisms which are holomorphic) namely, the group of Möbius transformations. It contains the complex plane whose automorphism group (the similarity group) is a subgroup of the Möbius group.

Holomorphic (meromorphic) functions on a Riemann surface are those continuous functions which viewed through the charts are holomorphic (meromorphic). We denote the field of meromorphic functions defined on a Riemann surface X by $\mathcal{M}(X)$. Meromorphic functions can be seen as holomorphic functions with values in $\mathbb{C}P^1$.

We will show later that meromorphic functions on $\mathbb{C}P^1$ are very simple to describe. They are all rational:

Definition 1.4 *A rational function $f : \mathbb{C}P^1 \rightarrow \mathbb{C}P^1$ is a function of the form*

$$f(z) = \frac{p(z)}{q(z)}$$

where $p(z)$ and $q(z)$ are polynomials.

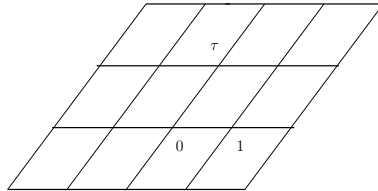
Writing a meromorphic function in the neighborhood of a point a as $f(z) = (z - a)^k g(z)$ where $g(a) \neq 0$, we define the order of $f(z)$ at a to be k . The order of the function defined on $\mathbb{C}P^1$ at ∞ is computed using the chart $w = 1/z$. So that if a $p(z)$ has degree n and $q(z)$ degree m , then ∞ will have order $-(n - m)$. Therefore, for a rational function, the sum of the orders of the zeros and poles is zero. Conversely, an easy construction gives

Proposition 1.5 *Let a_i and b_j be two finite disjoint families of points in $\mathbb{C}P^1$ with the same number of elements. Then there exists a rational function vanishing precisely at a_i and having poles precisely at b_j . The order of the function at each point being the number of times the point appears in the families.*

In contrast to the case of the Riemann sphere, this is not true for other Riemann surfaces. First of all, it is much more difficult to prove that there exists a meromorphic function and secondly one cannot fix arbitrarily the structure of zeros and poles of a meromorphic function.

The next examples of compact Riemann surfaces, after $\mathbb{C}P^1$, consists of complex structures on a torus. We will show later that any such structure arises as a quotient of $\mathbb{C}$ by a translation group generated by two independent directions one of each we may suppose (by a conjugation by a similarity transformation $z \rightarrow az + b$) to be $z \rightarrow z + 1$ and the other one $z \rightarrow z + \tau$ with $\tau \in \mathbb{C}$. More precisely, we will show that any compact Riemann surface whose underlying manifold is a torus is biholomorphic to an elliptic curve:

Definition 1.6 *Let $\tau \in \{ z \in \mathbb{C} \mid \text{Im } z > 0 \}$ and $\Gamma_\tau = \mathbb{Z} + \mathbb{Z}\tau$ be the additive group generated by $1, \tau \in \mathbb{C}$. We say that $E_\tau = \mathbb{C}/\Gamma_\tau$ is the complex torus associated to Γ_τ .*



The set of points inside the parallelogram defined by 1 and τ is called a fundamental region. Its closure, with some identifications on the boundary, is identified with a torus. Observe that any translation of that parallelogram also is a fundamental domain in the sense that any two points in its interior are contained in different orbits and each orbit has a point in the domain or its closure.

Meromorphic functions $\mathcal{M}(E_\tau)$ defined on E_τ are identified with meromorphic functions defined on $\mathbb{C}$ which are invariant under Γ_τ (called elliptic functions) but holomorphic functions which are invariant reduce to constants due to the maximum principle. It is not obvious that such a non-constant function exists but several of its properties, assuming existence, are simple to state. The following is a basic property.

Proposition 1.7 *Let $f \in \mathcal{M}(E_\tau)$ be a meromorphic function without poles on the boundary of a fundamental region. Then, the sum of its residues in the fundamental region is zero.*

Proof. The sum of residues in the interior is given by $\frac{1}{2\pi i} \int_{\partial P} f(z) dz$ where P is a parallelogram which is a fundamental domain. By translation invariance the integrals on opposite sides cancel. $\square$

This shows that, in order to construct a meromorphic function on E_τ with only one pole, its order has to be at least two. A related proposition counts the number of zeros.

Proposition 1.8 *Suppose there are no poles or zeros in the boundary of a fundamental domain. Then the number of zeros is the same as the number of poles counting multiplicities.*

Proof. The proof is simply a corollary to the previous proposition applied to the function f'/f . In fact the sum of the residues is equal to the number of zeros minus the number of poles counting multiplicity by the following

Exercise 1.9 *Prove that*

$$\frac{1}{2\pi i} \int_{\partial P} \frac{f'(z)}{f(z)} dz = \text{number of zeros} - \text{number of poles}$$

where P is the fundamental domain. $\square$

Another necessary condition on the zeros and poles of a meromorphic is given in the following proposition. It turns out that these necessary conditions are also sufficient (Abel's theorem).

Proposition 1.10 *Suppose there are no poles or zeros in the boundary of a fundamental domain. Let a_i and b_j be two finite disjoint families of points in a fundamental*

domain P and f an elliptic function vanishing precisely at a_i and having poles precisely at b_j (we repeat the points according to the multiplicity of the zero or pole). Then

$$\sum a_i - \sum b_j \in \Gamma_\tau.$$

Proof. The same as above using the following

Exercise 1.11

$$\frac{1}{2\pi i} \int_{\partial P} \frac{zf'(z)}{f(z)} dz = \sum a_i - \sum b_i.$$

Indeed, taking into account the invariance of $f(z)$ under translations and supposing that P is the parallelogram with corners $0, 1, 1 + \tau, \tau$, we get

$$\begin{aligned} \frac{1}{2\pi i} \int_{\partial P} \frac{zf'(z)}{f(z)} dz &= \frac{1}{2\pi i} \int_0^1 \frac{(-\tau)f'(z)}{f(z)} dz + \frac{1}{2\pi i} \int_0^\tau \frac{f'(z)}{f(z)} dz \\ &= \frac{1}{2\pi i} \left(-\tau \log(f(z)) \Big|_0^1 - \log(f(z)) \Big|_0^\tau \right) = n_1\tau + n_2. \end{aligned}$$

□

The next goal is to construct at least one meromorphic function on E_τ . In the following discussion we fix a translation τ and let Γ_τ be the lattice generated by 1 and τ . Several objects will depend on τ although we will not make it explicit. A direct construction of elliptic functions is obtained by means of the series

$$F_n(z) = \sum_{\gamma \in \Gamma_\tau} \frac{1}{(z - \gamma)^n}.$$

One can prove that the series converges absolutely and uniformly on compact sets for $n \geq 3$ so that $F_n(z)$ is meromorphic. To see that, we start with the following

Lemma 1.12 *The series*

$$\sum_{\gamma \in \Gamma_\tau - \{0\}} \frac{1}{|\gamma|^s}$$

is convergent for $s > 2$.

Proof. Consider the description of the lattice by the layers $n_1 + n_2\tau \in \Gamma_\tau$ with $\max(|n_1|, |n_2|) = n$. There are $8n$ elements of Γ_τ in that layer. If we let $r = \min(1, |\tau|, |1 + \tau|, |1 - \tau|)$ (that is the minimum of the norms of elements in the first layer), then $|n_1 + n_2\tau| \geq r \max(|n_1|, |n_2|)$. Therefore

$$\sum_{\gamma \in \Gamma_\tau - \{0\}} \frac{1}{|\gamma|^s} \leq \sum_{n \geq 1} \frac{8n}{r^s n^s} = \sum_{n \geq 1} \frac{8}{r^s n^{s-1}}$$

which is convergent for $s > 2$.

□

Lemma 1.13 *The series*

$$\sum_{\gamma \in \Gamma_\tau} \frac{1}{(z - \gamma)^s}$$

is uniformly convergent on compact sets of $\mathbb{C} - \Gamma_\tau$ for any integer $s > 2$.

Proof. If $K \subset \mathbb{C}$ is a compact subset we can assume that, except for finitely many γ , $|\gamma| \geq 2|z|$ for $z \in K$. In that case $|z - \gamma| \geq |\gamma| - |z| \geq |\gamma| - \frac{|\gamma|}{2} = \frac{|\gamma|}{2}$. Therefore for all $z \in K$ and γ on the complement of a finite subset in Γ_τ ,

$$\sum_{\gamma} \frac{1}{|z - \gamma|^s} \leq \sum_{\gamma} \frac{2^s}{|\gamma|^s}$$

which is convergent for $s > 2$. Together with the previous lemma, this implies the series is uniformly convergent by Weierstrass M -test. $\square$

Having proved convergence, for each $\omega \in \Gamma_\tau$ we obtain

$$F_n(z + \omega) = \sum_{\gamma \in \Gamma_\tau} \frac{1}{(z + \omega - \gamma)^n} = \sum_{\gamma \in \Gamma_\tau} \frac{1}{(z - \gamma)^n}$$

so that $F_n(z)$ is elliptic. In particular the function $F_3(z)$ is elliptic. It has a pole of order 3 at 0. To obtain a meromorphic function with a pole of order 2 we solve the equation

$$\mathcal{P}'(z) = -2F_3(z).$$

A solution is given by the Weierstrass function

$$\mathcal{P}(z) = \frac{1}{z^2} + \sum_{\gamma \in \Gamma_\tau - \{0\}} \left(\frac{1}{(z - \gamma)^2} - \frac{1}{\gamma^2} \right).$$

The naive idea would be to start with a function with a pole of order two, namely $\frac{1}{z^2}$, and add the term $\sum_{\gamma \in \Gamma_\tau - \{0\}} \frac{1}{(z - \gamma)^2}$ which would make it invariant under Γ_τ but unfortunately this sum is not convergent.

Lemma 1.14 *The series*

$$\mathcal{P}(z) = \frac{1}{z^2} + \sum_{\gamma \in \Gamma_\tau - \{0\}} \left(\frac{1}{(z - \gamma)^2} - \frac{1}{\gamma^2} \right).$$

defines an elliptic function with only one pole of order two modulo the lattice.

Proof. To show convergence, the argument is the same as in the previous lemma. The general term of the series $\mathcal{P}$ satisfies, as in the previous lemma, for $|\gamma| \geq 2|z|$

$$\left| \frac{1}{(z - \gamma)^2} - \frac{1}{\gamma^2} \right| = \left| \frac{z(z - 2\gamma)}{\gamma^2(z - \gamma)^2} \right| \leq \frac{4|z|(5/2|\gamma|)}{|\gamma|^2|\gamma|^2} \leq \frac{10|z|}{|\gamma|^3}.$$

Therefore, as before, we conclude that the series converges absolutely and uniformly on compact sets.

The periodicity is not clear from the formula. But we can use the periodicity of its derivative to conclude that $\mathcal{P}(z) - \mathcal{P}(z + 1)$ and $\mathcal{P}(z) - \mathcal{P}(z + \tau)$ are constants. The value of the constants are seen to be zero. In fact, $\mathcal{P}(-1/2) - \mathcal{P}(-1/2 + 1) = 0$ and $\mathcal{P}(-\tau/2) - \mathcal{P}(-\tau/2 + \tau) = 0$ because $\mathcal{P}(z)$ is clearly even. $\square$

The following existence theorem of meromorphic functions on an elliptic curve should be contrasted to the corresponding existence theorem of rational functions on the Riemann sphere. The only if part was proven in a previous proposition.

Theorem 1.15 (*Abel's theorem*) *Let E_τ be a complex torus with corresponding group Γ_τ . Let a_i and b_j be two finite disjoint families of points in a fundamental domain with the same number of elements (greater or equal than 2). Then there exists an elliptic function (restricted to the fundamental domain) vanishing precisely at a_i and having poles precisely at b_j if and only if*

$$\sum a_i - \sum b_j \in \Gamma_\tau.$$

Proof. (sketch) A constructive proof of this theorem can be given by considering the Weierstrass sigma functions (they are examples of theta functions)

$$\sigma(z) = z \prod_{\gamma \in \Gamma'} \left(1 - \frac{z}{\gamma}\right) e^{\frac{z}{\gamma} + \frac{z^2}{2\gamma^2}},$$

which have only simple zeros at points of Γ . They are not functions defined at the quotient but their behavior with respect to the lattice is quite simple. In fact

$$\sigma(z + \gamma) = (-1)^{n_\gamma} \sigma(z) e^{\alpha_\gamma(z + \frac{1}{2}\gamma)}$$

where α_γ and n_γ depend only on $\gamma \in \Gamma$. We define the meromorphic function in the theorem as

$$f(z) = \frac{\sigma(z - a_1) \cdots \sigma(z - a_n)}{\sigma(z - b_1) \cdots \sigma(z - b_n)}.$$

It is easy to verify that $f(z)$ is indeed defined on the quotient. $\square$

Another description of the set of meromorphic functions is given through its subsets of meromorphic functions with less singularities than a fixed arbitrary data. More precisely we fix a divisor, that is, a formal linear combination

$$D = \sum_z n_z [z]$$

where $n_z \in \mathbb{Z}$ are different from zero only for a finite number of $z \in E$. We think of a divisor as giving the order n_z of a possible function at z , except that a function with precisely these orders might not exist. The degree of a divisor will be the total order $\deg(D) = \sum_z n_z$. In particular we call divisor of f the divisor (called a principal divisor)

$$\operatorname{div}(f) = \sum_{z \in E} \operatorname{ord}_z(f) [z],$$

and we will show that it has zero degree for any meromorphic function defined on any compact Riemann surface. Define the vector space

$$L(D) = \{f \in \mathcal{M}(E) \mid f = 0 \text{ or } \operatorname{div}(f) \geq -D\}$$

where $\operatorname{div}(f) \geq -D$ means that, for each $z \in E$, the order of f at z is greater than or equal to $-n_z$. We will prove later the following:

Theorem 1.16 (*Riemann-Roch*) *Let D be a divisor on E with $d = \deg(D) \geq 1$. Then $\dim(L(D)) = d$.*

The field of meromorphic functions is described in the following

Theorem 1.17 $\mathcal{M}(E_\tau) = \mathbb{C}(P, P')$, *that is, the field of meromorphic functions is generated by $\mathbb{C}$, the Weierstrass function and its derivative.*

A meromorphic function on the elliptic curve can be interpreted as a function $E_\tau \rightarrow \mathbb{CP}^1$. In general, the meromorphic function is locally a bijection but it has ramification points when its derivatives vanishes. It is important then to determine the zeros of $\mathcal{P}'$:

Lemma 1.18 *The zeros of $\mathcal{P}'$ in a fundamental parallelogram containing the points $0, \frac{1}{2}, \frac{\tau}{2}$ and $\frac{1+\tau}{2}$ are*

$$\frac{1}{2}, \frac{\tau}{2}, \frac{1+\tau}{2}$$

Proof. As $\mathcal{P}'$ has order 3, it has only three zeros in the fundamental domain. We have $\mathcal{P}'(z) = -\mathcal{P}'(-z)$ because $\mathcal{P}'$ is odd. On the other hand, because $\mathcal{P}'$ is periodic, $\mathcal{P}'(z) = \mathcal{P}'(z - \gamma)$. Therefore, for $z = \gamma/2$, $\mathcal{P}'$ vanishes. $\square$

One can prove that the Weierstrass function defined on E_τ assumes each value on the Riemann sphere exactly twice except for 4 points; three corresponding to the vanishing of its derivative $\mathcal{P}(\frac{1}{2})$, $\mathcal{P}(\frac{\tau}{2})$, $\mathcal{P}(\frac{1+\tau}{2})$ and the last one corresponding to the unique pole of order 2, ∞ . That gives an interpretation of the Weierstrass function as a branched covering of the Riemann sphere by the torus.

Another way to obtain this interpretation is via the study of a differential equation satisfied by $\mathcal{P}(z)$ which, in fact, establishes an algebraic relation between $\mathcal{P}(z)$ and $\mathcal{P}'(z)$.

Proposition 1.19 *The Weierstrass function satisfies the equation*

$$\mathcal{P}'(z)^2 = 4\mathcal{P}^3(z) - g_2(\tau)\mathcal{P}(z) - g_3(\tau)$$

where

$$g_2(\tau) = 60 \sum_{\gamma \in \Gamma_\tau - \{0\}} \frac{1}{\gamma^4}$$

and

$$g_3(\tau) = 140 \sum_{\gamma \in \Gamma_\tau - \{0\}} \frac{1}{\gamma^6}.$$

Proof. A simple proof can be given by computing the Laurent series at the origin of $\mathcal{P}(z)$ and $\mathcal{P}'(z)$. One must show that the two sides of the equality have equal Laurent series up to the constant term. In that case their difference would be a bounded holomorphic function vanishing at the origin and therefore, by Liouville, vanishing everywhere.

In order to obtain the Laurent series of $\mathcal{P}(z)$ it is useful to consider the series below satisfying $\zeta'(z) = -\mathcal{P}(z)$.

$$\zeta(z) = \frac{1}{z} + \sum_{\gamma \in \Gamma_\tau - \{0\}} \left(\frac{1}{(z - \gamma)} + \frac{1}{\gamma} + \frac{z}{\gamma^2} \right)$$

Exercise 1.20 *The series is uniformly convergent and its Laurent series is*

$$\zeta(z) = \frac{1}{z} - G_4 z^3 - G_6 z^5 + \dots$$

where

$$G_n = \sum_{\gamma \in \Gamma_\tau - \{0\}} \frac{1}{\gamma^n}.$$

We obtain the following developments

$$\mathcal{P}(z) = -\zeta'(z) = \frac{1}{z^2} + 3G_4 z^2 + 5G_6 z^4 + \dots$$

$$4\mathcal{P}(z)^3 = \frac{4}{z^6} - \frac{36G_4}{z^2} - 60G_6 + \dots$$

$$\mathcal{P}'(z) = -\frac{2}{z^3} + 6G_4 z + 20G_6 z^3 + \dots$$

$$\mathcal{P}'(z)^2 = \frac{4}{z^6} - \frac{24G_4}{z^2} - 80G_6 + \dots$$

and then a simple computation shows that the Laurent series of each side of the equation is equal up to zero order. $\square$

Writing $t = \mathcal{P}(z)$ and the differential equation as $\left(\frac{dt}{dz}\right)^2 = 4t^3 - g_2 t - g_3$ we see that the inverse function of $\mathcal{P}(z)$, $\mathcal{P}^{-1}(t)$, is formally given by

$$\int \frac{1}{\sqrt{4t^3 - g_2 t - g_3}} dt.$$

But those integrals are not well defined in general. The problem is that the function $\sqrt{4t^3 - g_2 t - g_3}$ is not well defined in $\mathbb{C}$. For each path of integration (which does not meet the roots) one can define the integral by analytically extending the function along the path, but different paths will give, sometimes, different integrals.

In fact, the study of integrals of the form

$$\int \frac{1}{\sqrt{p(t)}} dt$$

were the motivation for the whole theory. In particular one can think of the elliptic functions as generalizations of the circular functions. For instance

$$\int \frac{1}{\sqrt{1-t^2}} dt$$

is $\arcsin(t)$ and the inverse function of that integral is a periodic function. The elliptic functions are inverse functions of the integrals as above with $p(t)$ of degree three and they have the remarkable property of being doubly periodic.

The map $E_\tau - [0] \rightarrow \mathbb{C}^2$ given by $z \rightarrow (\mathcal{P}(z), \mathcal{P}'(z))$ defined on the complement of the pole ($[0]$ is the projection of the lattice on the quotient space) is a holomorphic embedding whose image is the curve

$$y^2 = 4x^3 - g_2(\tau)x - g_3(\tau).$$

But one can extend that embedding to complex projective space.

Theorem 1.21 *The map $z \rightarrow (\mathcal{P}(z), \mathcal{P}'(z), 1)$ for $z \in \mathbb{C} - \Gamma_\tau$ and $z \rightarrow (0, 1, 0)$ for $z \in \Gamma_\tau$ defines a holomorphic embedding $E_\tau \rightarrow \mathbb{C}P^2$ whose image is the algebraic curve*

$$y^2 z = 4x^3 - g_2(\tau)xz^2 - g_3(\tau)z^3.$$

Several results about elliptic curves are generalized for any compact Riemann surface. In particular, we will

1. Describe any Riemann surface as a quotient of $\mathbb{C}$, D , the unit disc, or the Riemann sphere by a discrete group Γ .
2. Prove that there exist meromorphic functions on any compact surface and, more generally, give a generalization of Abel's theorem, Riemann-Roch theorem and describe the structure of its field of meromorphic functions.
3. Prove that there exists an embedding of a compact Riemann surface as a submanifold of a complex projective space.

2 Topology of surfaces

A two dimensional topological manifold is called a surface. That is a connected Hausdorff topological space M having a cover by open sets U_α and a collection of homeomorphisms $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^2$ which are compatible in the sense that the transition functions

$$\varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \rightarrow \varphi_\alpha(U_\alpha \cap U_\beta)$$

are homeomorphisms.

It is convenient to have a combinatorial description of surfaces by means of a triangulation. This allows a direct computation of some topological invariants of the surface as the Euler characteristic and the fundamental group.

To be more precise define first the standard 2-simplex Δ given by the convex envelope of the points (vertices) $(0, 0), (1, 0), (0, 1)$ in $\mathbb{R}^2$. Each boundary segment is called an edge. If $\varphi : \Delta \rightarrow \varphi(\Delta) \subset M$ is a homeomorphism, we call $\varphi(\Delta)$ a triangle and the images of the vertices and edges of the standard simplex are also called vertices and edges of the triangle.

Definition 2.1 *A triangulation of a compact surface M is a finite set of homeomorphisms $\varphi_i : \Delta \rightarrow \varphi_i(\Delta) \subset M$ such that the intersection of two triangles is either*

- *empty,*
- *a vertex or*
- *an edge of each of the triangles.*

In particular the interior of the triangles are disjoint. We can now state the theorem whose first rigorous proof was given by Radó in 1924.

Theorem 2.2 *Any compact surface has a triangulation.*

Remark 2.3 1. *In fact, Radó proved that any surface which has a countable basis of open sets can be triangulated. For non-compact surfaces, as the number of triangles is not finite, we need to impose that each point has a neighborhood intersecting only a finite number of triangles.*

2. *The existence of a triangulation for a compact manifold dimension 3 was established by Moise in 1952, but in dimensions higher than three a topological manifold might not have a triangulation.*
3. *One can define orientability for triangulated surfaces by saying that there exists a compatible orientation on all triangles (they induce the same orientation on common edges).*
4. *Any triangulation of a compact surface may be obtained from another one by a finite sequence of the following elementary moves:*
 - *the creation of a vertex inside a triangle and thereby introducing three new triangles in the place of the original one and the corresponding inverse operation,*
 - *replacing the common side of two adjacent triangles of the triangulation by the other diagonal of the quadrilateral formed by these two triangles (this is called a flip).*

A reference for the classification of compact surfaces is the first chapter of [Massey] and we will only summarize the main results below without proofs. Riemann surfaces are orientable surfaces because the transition functions, being holomorphic, preserve orientation so we state the theorem of classification only for orientable surfaces. A basic surgery construction is that of connected sum. We start with two surfaces and

remove one disc from each and glue the two surfaces along the boundary of the discs. In fact we can obtain any surface, apart the sphere, by this surgery procedure applied to tori.

Theorem 2.4 *A compact orientable surface is homeomorphic to a sphere or to a connected sum of tori.*

Proof. Sketch: Once we know the surface is triangulated, one can prove the theorem of classification of compact surfaces by spreading the triangulation of the surface in the plane to form a polygon with boundary identifications. More precisely, given a triangulated surface we enumerate its triangles $T_1, T_2, \dots, T_n$ in a way that each T_i has an edge in common with one of the previous triangles in the sequence. If T_i has two edges in common, we choose one of them to identify to one of the edges on the plane but leave the other one as a boundary of the polygon thus obtained. The union of the first two triangles along the common edge gives a parallelogram with possible boundary identifications. Adding each triangle makes the number of sides of this polygon jump by two. At the end we obtain a polygon with a number of sides identifications.

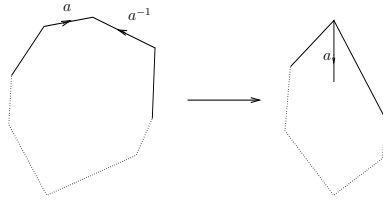
The idea now is to find a normal form for this polygon describing the surface. A usual normal form is the one which describes the surface as a connected sum of tori. A torus corresponds to a sequence $aba^{-1}b^{-1}$ and a handle to sequence $aba^{-1}b^{-1}c$. Clearly $a_1b_1a_1^{-1}b_1^{-1}a_2b_2a_2^{-1}b_2^{-1}$ corresponds to a connected sum of two tori. In other words, adding a handle to a torus. The normal form we look for a surface with g handles is therefore

$$a_1b_1a_1^{-1}b_1^{-1} \dots a_gb_ga_g^{-1}b_g^{-1}$$

as in Figure 1) with $g \geq 1$ or aa^{-1} which is a sphere.

This is done using a sequence of operations which simplify the structure of the identifications on the boundary. As a first observation, note that a sequence of edges of the form $a \dots a \dots$ cannot occur because this implies the existence of a Möbius band inside the surface which would not be orientable. We divide the proof in several steps which are better explained by the figures:

1. Elimination of adjacent edges of the form aa^{-1} (in the case there are more than two sides).



2. Obtain a polygon whose vertices are all identified. In the figure we show how to get rid of a vertex P at the expense of introducing more vertices identified to Q . Each time we perform this operation we return to step one to simplify the configuration. If there is only one point P in the boundary, then we obtain

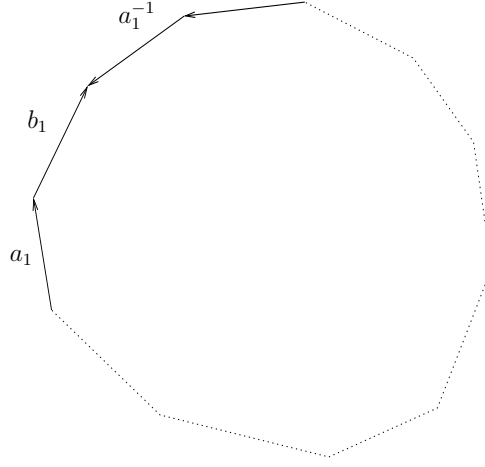


Figure 1: A surface obtained by boundary identifications on a disc.

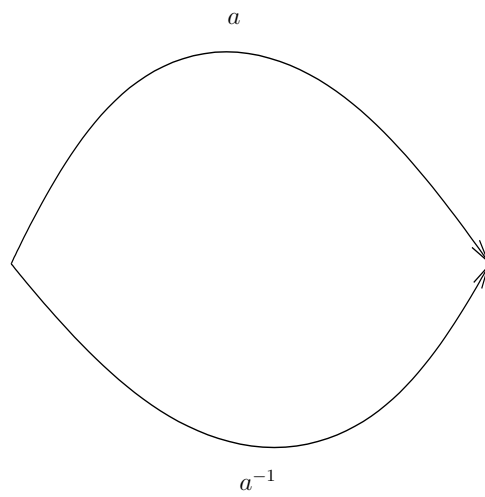
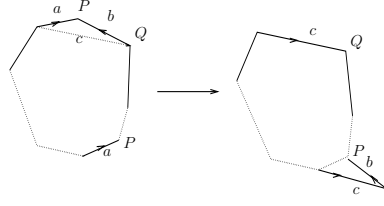


Figure 2: A sphere obtained by boundary identifications on a disc.

a configuration as in step one so either we obtain a sphere or we can simplify, therefore suppressing the point P .



3. Change to a sequence of handles: exercise.

□

The Euler characteristic can be defined by the formula $\chi = T - E + V$, where T is the number of triangles, E is the number of edges and V is the number of vertices of a triangulation. It is given by

$$\chi = 2 - 2g.$$

3 Covering spaces and the fundamental group

A curve in a topological space X is a continuous map $c : [0, 1] \rightarrow X$. Two curves c_1 and c_2 with $c_1(0) = c_2(0)$ and $c_1(1) = c_2(1)$ are homotopic (with fixed end points) if there exists a continuous map $F : [0, 1] \times [0, 1] \rightarrow X$ such that

1. $F|_{\{0\} \times [0,1]} = c_1(0)$ and $F|_{\{1\} \times [0,1]} = c_1(1)$
2. $F|_{[0,1] \times \{0\}} = c_1$ and $F|_{[0,1] \times \{1\}} = c_2$.

A loop in X is a curve c with $c(0) = c(1)$. We can define the product of two loops c_1 and c_2 as the loop $c_2c_1 : [0, 1] \rightarrow X$ given by $c_2c_1(t) = c_1(2t)$ for $0 \leq t \leq 1/2$ and $c_2c_1(t) = c_2(2(t - 1/2))$ for $1/2 \leq t \leq 1$. The constant loop is defined to be $c(t) = x_0$ for all t , and the inverse of a loop c is the loop c^{-1} defined by $c^{-1}(t) = c(1 - t)$. Two loops c_1 and c_2 with $c_1(0) = c_2(0) = x_0$ (we say the loops are based at x_0) are homotopic if there exists a continuous map $F : [0, 1] \times [0, 1] \rightarrow X$ such that

1. $F|_{\{0\} \times [0,1]} = F|_{\{1\} \times [0,1]} = x_0$
2. $F|_{[0,1] \times \{0\}} = c_1$ and $F|_{[0,1] \times \{1\}} = c_2$

We say that two loops are freely homotopic if the second condition is verified but not necessarily the first one.

Let X be a manifold and $x_0 \in X$ a base point. We denote by $\pi_1(X, x_0)$ the space of homotopy classes of loops based at x_0 . It has a group structure induced by the multiplication on loops. Usually we denote by $[\gamma]$ the class containing the loop γ .

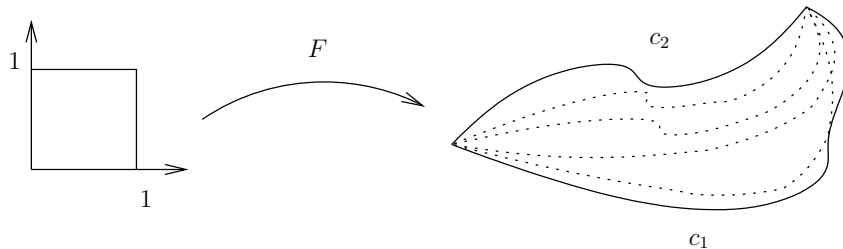


Figure 3: A homotopy between two curves c_1 and c_2 .

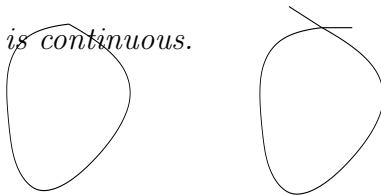
If x'_0 is another base point, $\pi_1(X, x'_0)$ is isomorphic to $\pi_1(X, x_0)$. In fact, let c be a curve with $c(0) = x_0$ and $c(1) = x'_0$. Then, one can define $\pi_1(X, x_0) \rightarrow \pi_1(X, x'_0)$ by $\gamma \rightarrow c\gamma c^{-1}$.

Example 3.1 *The fundamental group of S^1 is isomorphic to $\mathbb{Z}$.*

A continuous function $f : X \rightarrow Y$ between topological spaces such that $f(x_0) = y_0$ induces a homomorphism $f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$. A homeomorphism induces an isomorphism but an isomorphism between fundamental groups does not imply that the corresponding topological spaces are homeomorphic. A typical situation of isomorphic fundamental groups arises in the case of deformation retracts. They are very useful for computations.

Definition 3.2 *A subset $K \subset X$ of a topological space is a deformation retract of X if there exists a homotopy $F : X \times [0, 1] \rightarrow X$ such that*

- *For all $x \in X$, $F(x, 0) = x$.*
- *For all $x \in K$, $F(x, \cdot) = x$.*
- *$F(\cdot, 1) : X \rightarrow K$ is continuous.*



3.1 Group presentations and computations of the fundamental group.

A presentation of a group Γ is given by

$$\Gamma = \langle \gamma_1, \dots \mid r_1, \dots \rangle.$$

The γ_i are the generators and the r_i reduced words on the generators (words constructed with γ_i or γ_i^{-1} which don't contain the sequence $\gamma_i\gamma_i^{-1}$). By definition Γ is isomorphic to the quotient of the free group on the generators γ_i by the normal subgroup generated by the relators. We say that Γ is finitely presented if there exists a presentation with a finitely number of generators and relators.

Example 3.3

$$\mathbb{Z} \oplus \mathbb{Z} = \langle \gamma_1, \gamma_2 \mid [\gamma_1, \gamma_2] \rangle.$$

To give the fundamental group by a presentation is very useful for computations. An application of that description is the following theorem which we quote without proof.

Theorem 3.4 (Seifert-Van Kampen Theorem) *Let $M = M_1 \cup M_2$, two path-connected open sets with $I = M_1 \cap M_2$ path-connected. Suppose the fundamental groups of M_1 and M_2 at a base point $x_0 \in I$ are $\Gamma_1 = \langle \gamma_1, \dots \mid r_1, \dots \rangle$ and $\Gamma_2 = \langle \delta_1, \dots \mid s_1, \dots \rangle$. Suppose $\pi_1(I, x_0)$ is generated by generators η_i . Write, for each η_i , the corresponding word φ_i and ψ_i in the generators of Γ_i . Then*

$$\pi_1(M, x_0) = \langle \gamma_1, \dots, \delta_1, \dots \mid r_1, \dots, s_1, \dots, \varphi_i \psi_i^{-1} \rangle.$$

As a first application of the theorem we compute

Exercise 3.5 *The fundamental group of the infinity symbol ∞ is the free group with two generators. More generally, the fundamental group of a bouquet of g circles is the free group with g generators.*

We use the theorem of Seifert-Van Kampen to provide presentations for surface groups.

Exercise 3.6 *The fundamental group of a compact Riemann surface of genus g with a point deleted is the free group with $2g$ generators.*

We say a surface is of finite type if it is homeomorphic to a compact surface with a finite number of points (or disjoint discs) deleted.

Theorem 3.7 *The fundamental group of an orientable surface of finite type has a presentation of the form*

$$\langle a_1, b_1, \dots, a_g, b_g, h_1, \dots, h_t \mid \prod_{j=1}^g [a_j, b_j] h_1 \cdots h_t = 1 \rangle.$$

The elements h_i correspond to loops around the boundaries. In particular, from the presentation, we see that if $t \neq 0$ the fundamental group is free of rank $2g + t - 1$.

3.2 Covering spaces

In the following we suppose that the topological spaces are all arc connected and locally arc connected. We denote by $\varphi : (Y, y_0) \rightarrow (X, x_0)$ a continuous map $\varphi : Y \rightarrow X$ such that $\varphi(y_0) = x_0$. Recall that it induces the homomorphism $\varphi_* : \pi_1(Y, y_0) \rightarrow \pi_1(X, x_0)$ defined by $[\gamma] \rightarrow [\varphi \circ \gamma]$.

Definition 3.8 *A map $p : \tilde{X} \rightarrow X$ between topological spaces is a covering if each point $x \in X$ has a neighborhood U_x such that $p^{-1}(U_x)$ is a disjoint union of open sets homeomorphic to U_x under p .*

We say that two coverings $p_1 : \tilde{X}_1 \rightarrow X$ and $p_2 : \tilde{X}_2 \rightarrow X$ are equivalent if there exists a homeomorphism $p : \tilde{X}_1 \rightarrow \tilde{X}_2$ such that $p_2 \circ p = p_1$. Coverings have the fundamental path lifting property:

Proposition 3.9 *Let $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a covering space. A path $\varphi : ([0, 1], 0) \rightarrow (X, x_0)$ can be lifted to a unique path $\tilde{\varphi} : ([0, 1], 0) \rightarrow (\tilde{X}, \tilde{x}_0)$ satisfying $p \circ \tilde{\varphi} = \varphi$.*

Proof. Let $L = \{ t \in [0, 1] \mid [0, t] \text{ can be lifted} \}$. We will show that this set is open and closed. It is clearly non-empty as $0 \in L$. If $t_0 \in L$ then $\tilde{\varphi}(t_0)$ is contained in a unique component U of $p^{-1}(V)$ homeomorphic to V , a sufficiently small neighborhood of $\varphi(t_0)$. There exists therefore a lift of the curve in a neighborhood of t_0 by taking $(p|_U)^{-1} \circ \varphi$. Similarly if t_0 is a limit of points t_n in L we observe that there exists a sufficiently small neighborhood of $\varphi(t_0)$ such that $\tilde{\varphi}(t_n)$ are contained in a component U of $p^{-1}(V)$. As U is a homeomorphism we can define $\tilde{\varphi}(t_0)$. Uniqueness follows by a similar argument. $\square$

Using a similar proof we may lift homotopies on X to homotopies on a covering $\tilde{X}$:

Proposition 3.10 *Let $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a covering space. A homotopy $F : [0, 1] \times [0, 1] \rightarrow X$ between two paths $\varphi_1 : ([0, 1], 0) \rightarrow (X, x_0)$ and $\varphi_2 : ([0, 1], 0) \rightarrow (X, x_0)$ has a lift to a unique homotopy $\tilde{F} : [0, 1] \times [0, 1] \rightarrow \tilde{X}$ between $\tilde{\varphi}_1 : ([0, 1], 0) \rightarrow (\tilde{X}, \tilde{x}_0)$ and $\tilde{\varphi}_2 : ([0, 1], 0) \rightarrow (\tilde{X}, \tilde{x}_0)$. In particular, $\tilde{\varphi}_1(1) = \tilde{\varphi}_2(1)$.*

Remark 3.11 1. *The proposition above shows that $p_*\pi_1(\tilde{X}, \tilde{x}_0) \rightarrow \pi_1(X, x_0)$ is injective.*

2. *If $\tilde{x}'_0$ is another base point for $\tilde{X}$ over x_0 then $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ and $p_*(\pi_1(\tilde{X}, \tilde{x}'_0))$ are conjugate.*

Definition 3.12 *The subgroup $p_*\pi_1(\tilde{X}, \tilde{x}_0) \subset \pi_1(X, x_0)$ is called the defining subgroup of the covering.*

Definition 3.13 *The universal covering of a topological space (arc connected and locally arc connected) is the covering having trivial defining group.*

The basic result about covering spaces is the following:

Theorem 3.14 *There exists a bijection between conjugacy classes of subgroups of $\pi_1(X, x_0)$ and equivalence classes of coverings.*

The construction of the covering space associated to a given subgroup $\Gamma \subset \pi_1(X, x_0)$ can be accomplished by considering the set of equivalence classes of paths $c : [0, 1] \rightarrow X$ with $c(0) = x_0$. Equivalence between paths c_1 and c_2 meaning that $c_1(1) = c_2(1)$ and that $[c_2^{-1}c_1] \in \Gamma$. The map $p : \tilde{X} \rightarrow X$ is given by $p([c]) = c(1)$. For details see [Massey].

Remark 3.15 *If X is simply connected any covering is homeomorphic to X .*

The covering transformations (or deck transformations) of a covering $p : \tilde{X} \rightarrow X$ are those homeomorphisms $\varphi : \tilde{X} \rightarrow \tilde{X}$ satisfying $\pi \circ \varphi = \pi$. The description of the covering group is given in the following theorem.

Theorem 3.16 *The group of covering transformations is isomorphic to*

$$N(p_*\pi_1(\tilde{X}, \tilde{x}_0))/p_*\pi_1(\tilde{X}, \tilde{x}_0)$$

where N denotes the normalizer of the group in $\pi_1(X, x_0)$.

A covering whose defining subgroup is normal is called a *regular* or *normal* covering. In particular the universal covering is regular and $\pi_1(X, x_0)$ is the group of covering transformations.

3.3 Exercises

1. Recall that a map $\varphi : X \rightarrow Y$ is proper if for any compact $K \subset Y$, $\varphi^{-1}(K)$ is compact. Show that a local homeomorphism between manifolds is a finite covering if and only if φ is proper.
2. The punctured unit disc D^* has the upper half-plane as a universal covering. An explicit map is given by $e^{2\pi it}$. The fundamental group is $\mathbb{Z}$ acting on the half-plane by integer translations. The regular covering corresponding to the subgroup generated by $e^{2\pi im}$ also is the disc with covering group isomorphic to $\mathbb{Z}/m\mathbb{Z}$. The finite coverings of the punctured unit disc are equivalent to the maps $\varphi_m : D^* \rightarrow D^*$ given by $z \rightarrow z^m$.
3. The torus $S^1 \times S^1$ is covered by the plane. Find its regular coverings.
4. Let the annulus $A = \{ r < |w| < 1 \}$. The map $z \rightarrow \exp(2\pi i \log z / \log \lambda)$, where $r = \exp(-2\pi^2 / \log \lambda)$ defines a covering $D \rightarrow A$ of A by the unit disc D . The covering group is generated by $z \rightarrow \lambda z$.
5. Give an example of a surjective map which is a local homeomorphism but which is not a covering.

6. Let X be a simply connected Riemann surface and $f : X \rightarrow \mathbb{C}^*$ a holomorphic function. Prove that there exists a function $\tilde{f} : X \rightarrow \mathbb{C}$ such that $\exp \circ \tilde{f} = f$.
7. Let M_1 and M_2 be two manifolds which have the same universal covering $\tilde{M}$ with projections $p_1 : \tilde{M} \rightarrow M_1$ and $p_2 : \tilde{M} \rightarrow M_2$ and covering transformations group G_1 and G_2 respectively. If $\varphi : M_1 \rightarrow M_2$ is a homeomorphism, then we can lift it to a homeomorphism $\tilde{\varphi} : \tilde{M} \rightarrow \tilde{M}$. Prove that $G_2 = \tilde{\varphi} \circ G_1 \circ \tilde{\varphi}$.

3.4 Group actions

Let G be a group and X a topological manifold.

Definition 3.17 G acts by homeomorphisms on X if there exists a map $G \times X \rightarrow X$ such that

1. for fixed $g \in G$, the induced map $g : X \rightarrow X$ is a homeomorphism.
2. $(gh)x = g(hx)$ for all $x \in X$ and $g, h \in G$
3. $1x = x$ for all $x \in X$

If $G \times X \rightarrow X$ is an action we call the set $G_x = \{g \in G \mid gx = x\}$ the *stabilizer* or *isotropy of the action at x* . The *orbit* of $x \in X$ is the set Gx . The action is said to be *transitive* if the orbit of every point coincides with the whole space. The set of all orbits is denoted X/G and we define a topology on it by imposing that $U \subset X/G$ is open if and only if $\pi^{-1}(U) \subset X$ is open, where $\pi : X \rightarrow X/G$ is the canonical projection. A very special action is related to covering spaces. We need the following definitions:

Definition 3.18 Let $G \times X \rightarrow X$ be an action.

1. The action of G is *free* if no point of X is fixed by an element of G different from the identity (that is, the isotropy of each element of X is trivial).
2. The action is *properly discontinuous* if for any compact $K \subset X$ the set of all $\gamma \in G$ such that $\gamma K \cap K \neq \emptyset$ is finite.

Proposition 3.19 Let $G \times X \rightarrow X$ be an action on a manifold X . The quotient X/G is a manifold with projection $X \rightarrow X/G$ a covering if and only if the action is free and properly discontinuous.

Proof. We will prove that the projection is a covering onto a manifold if the action is free and properly discontinuous leaving the other implication as an exercise. Suppose $x \in X$ and U_x is a relatively compact neighborhood. As the action is properly discontinuous there exists only a finite number of elements in G such that $g\bar{U}_x \cap \bar{U}_x \neq \emptyset$. As the action is free, for each one of those elements, $gx \neq x$. As the space is Hausdorff, we can choose a neighborhood $V_x \subset U_x$ such that for all $g \in G$, $g\bar{V}_x \cap \bar{V}_x = \emptyset$. This proves that the projection $X \rightarrow X/G$ is a covering.

The quotient is Hausdorff: suppose $x, y \in X$ are two points in distinct orbits. As X is a manifold, there exists two relatively compact neighborhoods U_x and U_y with $\bar{U}_x \cap \bar{U}_y = \emptyset$. As before, because the action is properly discontinuous and free, we may suppose $g\bar{U}_x \cap \bar{U}_x = \emptyset$ and $g\bar{U}_y \cap \bar{U}_y = \emptyset$. Consider $K = \bar{U}_x \cup \bar{U}_y$. As the action is properly discontinuous, the set of elements $g \in G$ such that $gK \cap K = (g\bar{U}_x \cap \bar{U}_y) \cup (\bar{U}_x \cap g\bar{U}_y) \neq \emptyset$ is finite, and by the same argument as before (using the fact that the action is free), we can choose U_x and U_y smaller such that $gK \cap K = \emptyset$ for all g . $\square$

Exercise 3.20 *A discrete subgroup Γ of a topological group G acts freely properly discontinuously on G by the natural action $\Gamma \times G \rightarrow G$ given by $(\gamma, g) \rightarrow \gamma g$.*

Example 3.21 *A subgroup of $\mathbb{R}^n$ is discrete if and only if it is generated by a set of linearly independent vectors.*

Proof. Suppose that the group is generated by a set of linearly independent vectors. By a linear transformation we can transform the set into a subset of the canonical base vectors. It is clear that the group is discrete as 0 is an isolated point of the group.

Conversely, suppose that the subgroup $\Gamma \subset \mathbb{R}^n$ is discrete and use induction on the dimension. For $n = 1$, let v be the smallest positive vector. Without loss of generality, suppose $\gamma \in \Gamma$ is positive and let k be the largest integer such that $kv \leq \gamma$. Then $\gamma - kv \in \Gamma$ and is smaller than v . A contradiction unless $\gamma = kv$. We conclude that Γ is generated by v .

Suppose now that any discrete subgroup in $\mathbb{R}^{n-1}$ is generated by a set of linearly independent vectors. Let $\Gamma \subset \mathbb{R}^n$ be discrete and v a vector with minimum norm. Because of the first step of the induction $\Gamma \cap \mathbb{R}v = \mathbb{Z}v$. Let $\pi : \mathbb{R}^n \rightarrow \mathbb{R}^n/\mathbb{R}v$ be the quotient map. We claim that $\pi(\Gamma)$ is discrete. Suppose v_i is a sequence in Γ such that $\pi(v_i) \rightarrow 0$, that is, $v_i - r_i v \rightarrow 0$ (where we can suppose that $r_i \leq 1/2$). Then for large i , $v_i < v$. This implies that $v_i = 0$ for large i so that $\pi(\Gamma)$ is discrete. By the induction hypothesis we can find linearly independent vectors $\{\pi(w_1), \dots, \pi(w_{m-1})\}$ generating $\pi(\Gamma)$. $\{v, w_1, \dots, w_{m-1}\}$ are linearly independent and generate Γ . $\square$

4 Branched coverings and holomorphic functions

Holomorphic and meromorphic functions $\varphi : X \rightarrow \mathbb{C}$ on a Riemann surface are defined to be continuous functions for which the composition $\varphi \circ p^{-1}$, where p^{-1} is any chart, is holomorphic or meromorphic.

The local behavior of a holomorphic map between Riemann surfaces is described in the following theorem.

Theorem 4.1 *Let $\varphi : Y \rightarrow X$ be a non-constant holomorphic map between Riemann surfaces with $\varphi(y_0) = x_0$. There exists local coordinates p_Y and p_X around y_0 and x_0 respectively such that $p_Y(y_0) = p_X(x_0) = 0$ and $p_X \circ \varphi \circ p_Y^{-1}(z) = z^n$ for some $n \geq 1$.*

Proof. Clearly we can assume that there exists local coordinates p'_Y (we will change that coordinate next) and p_X around y_0 and x_0 , respectively, such that $p_Y(y_0) = p_X(x_0) = 0$. Now, if $p_X \circ \varphi \circ p_Y'^{-1}$ is non-constant we may suppose that there exists a holomorphic function $f(w)$ such that $p_X \circ \varphi \circ p_Y'^{-1}(w) = w^n f(w)$ with $n \geq 1$ and $f(0) \neq 0$. Therefore, on some neighborhood of the origin, there exists a holomorphic function $h(w)$ such that $h^n(w) = f(w)$. Observe that the map $p : w \rightarrow wh(w)$ is a biholomorphism in a neighborhood of the origin so that $p_Y = p \circ p_Y'$ is a new chart around y_0 . For $z = wh(w)$ we obtain $p_X \circ \varphi \circ p_Y^{-1}(z) = p_X \circ \varphi \circ p_Y'^{-1}(w) = w^n f(w) = (wh(w))^n = z^n$. $\square$

Observe that in the case $n = 1$ the map φ is a local biholomorphism at $y_0 \in Y$.

Definition 4.2 *A point $y_0 \in Y$ with $n \geq 2$ in the above theorem is called a branch point or a ramification point of order n of the map φ .*

Definition 4.3 *A map $\varphi : Y \rightarrow X$ onto a surface is a branched covering if*

1. *The restriction $\varphi|_{\varphi^{-1}(X-S)}$, where S is a discrete subset of X , is a covering.*
2. *For each point in $y_0 \in \varphi^{-1}(S)$ there are coordinates p_Y around y_0 and p_X around $x_0 = \varphi(y_0)$ such that $p_X \circ \varphi \circ p_Y^{-1}(z) = z^n$.*

A finite covering of a Riemann surface with a number of points deleted can always be extended to a branched covering. More precisely:

Theorem 4.4 *If X is a Riemann surface and $S \subset X$ is a closed discrete subset, then any finite covering $\varphi' : Y' \rightarrow X' = X \setminus S$ can be extended to a proper holomorphic map $\varphi : Y \rightarrow X$, where Y is a Riemann surface containing Y' such that $Y \setminus Y'$ is a closed discrete subset.*

Proof. At a point $s \in S$ there exists a neighborhood U_s with $U_s \cap S = \{s\}$ and a coordinate chart $\varphi_s : U_s \rightarrow D$ where D is the unit disc centered at the origin. As φ' is a finite covering, there exists a finite number of components $\varphi'^{-1}(U_s \setminus \{s\})$. Let V' be one of the components. As $\varphi'|_{V'}$ is a finite covering of the unit punctured disc, there exists a map $\psi' : V' \rightarrow D \setminus \{0\}$ so that $\varphi_s \circ \varphi \circ \psi'^{-1} : D \setminus \{0\} \rightarrow D \setminus \{0\}$ such that $\varphi_s \circ \varphi \circ \psi'^{-1}(z) = z^k$ and therefore we can add the point 0 to $D \setminus \{0\}$ and obtain a holomorphic map from D to D . Let V be the set obtained by adding an abstract point to V' so that $\psi : V \rightarrow D$ is a homeomorphism and defines a holomorphic chart and $\varphi|_V$ becomes a branched holomorphic covering. Repeating the procedure for each component above every $U_s \setminus \{s\}$ for $s \in S$ we obtain the Riemann surface Y . $\square$

4.0.1 Exercises

1. Prove Liouville's theorem: every bounded holomorphic function defined on $\mathbb{C}$ is constant.
2. Let $\varphi : Y \rightarrow X$ be a non-constant holomorphic map between Riemann surfaces. Show that φ is an open map.
3. Let $\varphi : X \rightarrow \mathbb{C}$ be a non-constant holomorphic map. Show that $|\varphi|$ does not attain its maximum. Conclude that every holomorphic function on a compact Riemann surface is constant.
4. Let $\varphi : X \rightarrow \mathbb{C}$ be a non-constant holomorphic map. Show that $\operatorname{Re} \varphi$ does not attain its maximum.
5. Show that the meromorphic functions on $\mathbb{C}P^1$ are quotients of two polynomials.
6. Let $\varphi : Y \rightarrow X$ be a non-constant holomorphic map between compact Riemann surfaces. Show that φ is surjective. Prove the fundamental theorem of algebra by considering a polynomial as a holomorphic map between $\mathbb{C}P^1$.

4.1 Algebraic functions

Definition 4.5 *The field of meromorphic functions defined on a Riemann surface X is denoted by $\mathcal{M}(X)$.*

Let $\varphi : Y \rightarrow X$ be a branched holomorphic covering of degree n between Riemann surfaces. The map $\varphi^* : \mathcal{M}(X) \rightarrow \mathcal{M}(Y)$ defined by $g \mapsto g \circ \varphi$ is clearly a monomorphism. Considering the field extension $\varphi^*(\mathcal{M}(X)) \subset \mathcal{M}(Y)$ we show the following

Theorem 4.6 *Let $\varphi : Y \rightarrow X$ be a branched holomorphic covering of degree n between Riemann surfaces. Then $\varphi^*(\mathcal{M}(X)) \subset \mathcal{M}(Y)$ is an algebraic field extension of degree less than n .*

Proof. Let $f \in \mathcal{M}(Y)$. Let $S \subset X$ be a closed discrete subset such that $\varphi : Y \setminus \varphi^{-1}(S) \rightarrow X \setminus S$ is a covering. Consider the restriction of f to the meromorphic function $f \in \mathcal{M}(Y \setminus \varphi^{-1}(S))$. We can define meromorphic functions on $X \setminus S$ by taking the elementary symmetric functions $s_1, \dots, s_n$ of the n functions $f \circ f_i : U \rightarrow \mathbb{C}$ where $f_i = \varphi|_{U_i}^{-1}$ and U_i is a component of $\varphi^{-1}(U)$ (supposing that each component of $\varphi^{-1}(U)$ is homeomorphic to U). Observe that, by construction, f is a solution of the equation

$$\prod_{i=1}^n (w - \varphi^*(f \circ f_i)) = w^n - \varphi^* s_1 w^{n-1} + \dots + (-1)^n \varphi^* s_n = 0.$$

To conclude that the extension is algebraic we need to show that the coefficients s_i extend to meromorphic functions on X . We divide the proof in two steps:

1. If f is holomorphic then s_i are bounded holomorphic functions on a neighborhood of a point $s \in S$. By Riemann's removable singularity theorem we can extend s_i to a holomorphic function.
2. If f is meromorphic at all points in $\varphi^{-1}(s)$, consider a coordinate chart $z : U \rightarrow D$ such that $z(s) = 0$. Then $(\varphi^*z)^m f$ is holomorphic if m is large and therefore the elementary symmetric functions of $(\varphi^*z)^m f$ can be extended to holomorphic functions of the form $z^{m_i} s_i$ and therefore the s_i can be extended to meromorphic functions.

Suppose $f_0 \in \mathcal{M}(Y)$ is an element such that the minimal polynomial is of maximal degree n_0 . We show now that $\mathcal{M}(X)(f_0) = \mathcal{M}(Y)$, thereby proving that the degree of the extension is less than n . In fact if $f \in \mathcal{M}(Y)$ is another element we have, by the existence of a primitive element ($\mathcal{M}(X)$ is of characteristic 0), $\mathcal{M}(X)(f_0, f) = \mathcal{M}(X)(g)$ and then

$$n_0 = \dim_{\mathcal{M}(X)} \mathcal{M}(X)(f_0) \leq \dim_{\mathcal{M}(X)} \mathcal{M}(X)(f_0, f) = \dim_{\mathcal{M}(X)} \mathcal{M}(X)(g) \leq n_0$$

so that $\mathcal{M}(X)(f_0) = \mathcal{M}(X)(f_0, f)$. □

In fact the degree of the extension in the theorem above is precisely n but, to prove that, we need a result (see corollary ??) that guarantees the existence of a meromorphic function which assumes pairwise different values at points of a generic fiber (that is, the inverse image of a point which is not a ramification value).

In the following we will prove a converse to that theorem. One of origins of Riemann surface theory concerns the study of algebraic equations of the form

$$w^n + a_1(z)w^{n-1} + \cdots + a_n(z) = 0,$$

where the coefficients $a_i(z)$ are meromorphic functions on $\mathbb{C}$. The idea is that the solution to that equation is, in fact, defined on a Riemann surface Y which is a branched covering $Y \rightarrow \mathbb{C}P^1$. We state the theorem in a more general form substituting $\mathbb{C}$ for a general Riemann surface X .

Theorem 4.7 *Let X be a Riemann surface and*

$$P(w) = w^n + a_1 w^{n-1} + \cdots + a_n$$

an irreducible polynomial in $\mathcal{M}(X)[w]$ of degree n . Then there exists a Riemann surface Y , a branched holomorphic covering $p : Y \rightarrow X$ of degree n and a meromorphic function $F \in \mathcal{M}(Y)$ such that

$$P(F) = F^n + p^* a_1 F^{n-1} + \cdots + p^* a_n = 0.$$

Definition 4.8 *We say that Y is the Riemann surface associated to the irreducible polynomial P .*

Remark 4.9 1. As $\mathcal{M}(X)$ is a field of characteristic 0, we know that the irreducible polynomial $P(w) \in \mathcal{M}(X)[w]$ is separable. That is, its roots in the algebraic closure of $\mathcal{M}(X)$ are all distinct.

2. Recall that the elementary symmetric polynomial $s_i(t_1, \dots, t_n)$ ($1 \leq i \leq n$) of the variables t_i generate the algebra of symmetric polynomials of those variables. Observe that the functions $a_i \in \mathcal{M}(X)$ are the elementary symmetric functions of the roots of the polynomial $P(w)$. That is

$$\prod_{1 \leq i \leq n} (w - t_i) = w^n - s_1 w^{n-1} + \dots + (-1)^n s_n.$$

Therefore, the polynomial $\Delta = \prod_{i < j} (t_i - t_j)^2$ which is clearly symmetric belongs to $\mathcal{M}(X)$. It is called the discriminant of $P(w)$. In particular, by the previous remark, the discriminant vanishes identically only if $P(w)$ is reducible.

Proof. The discriminant Δ of $P(w)$ vanishes at points of X where there are multiple roots. Therefore, because $P(w)$ is irreducible, Δ vanishes only on a closed discrete set of points S which we also suppose contains the poles of a_i . Let $X' = X \setminus S$ and define Y' to be the set of all points in $(z, w) \in (X \setminus S) \times \mathbb{C}$ satisfying the equation $P(w) = 0$. By the implicit function theorem (Proposition 4.10 below) and its corollary, $\varphi' : Y' \rightarrow X'$ is a covering map. We extend then this covering to a branched covering $\varphi : Y \rightarrow X$. The meromorphic function is defined first as a holomorphic function on Y' as $(z, w) \rightarrow w$ and then by extension (with a similar argument as in the previous theorem) to the whole of Y . To show that Y is connected, suppose that $Y = Y_1 \cup \dots \cup Y_k$ is a decomposition in connected components with $\varphi_i : Y_i \rightarrow X$ branched coverings. Then, for each φ_i the meromorphic function F restricted to Y_i defines a polynomial $P_i(w) \in \mathcal{M}(X)$ such that $P(w) = P_1(w) \cdots P_k(w)$ contradicting the irreducibility of $P(w)$. $\square$

Proposition 4.10 Let f be a holomorphic function in two complex variables defined on $\{ (z, w) \mid |z| < \epsilon_1, |w| = \epsilon_2 \}$. Suppose that $f(0, 0) = 0$ and $\frac{\partial f}{\partial w}(0, 0) \neq 0$. Then, there exists $\delta_1, \delta_2 > 0$ and a unique function φ defined on $|z| < \delta_1$ with $|\varphi| < \delta_2$ such that $f(z, \varphi(z)) = 0$. Moreover, φ is holomorphic.

Proof. As $\frac{\partial f}{\partial w}(0, 0) \neq 0$, there exists $\delta_2 > 0$ such that $f(0, w) \neq 0$ for $|w| < \delta_2$. There exists therefore, by compactness, $\delta_1 > 0$ such that $f(z, w) \neq 0$ for $|z| < \delta_1, |w| = \delta_2$. Writing $f_w(z, w) = \frac{\partial f(z, w)}{\partial w}$, for each z , the number of zeros of $f(z, w)$ in $|w| < \delta_2$ is given by the holomorphic function

$$N(z) = \frac{1}{2\pi i} \int_{|w|=\delta_2} \frac{f_w(z, w)}{f(z, w)} dw$$

which is therefore constant equal to one. The explicit solution is given by the residue theorem (writing $f(z, w) = (w - \varphi(z))h(z, w)$ for a non-vanishing function $h(z, w)$):

$$\varphi(z) = \frac{1}{2\pi i} \int_{|w|=\delta_2} w \frac{f_w(z, w)}{f(z, w)} dw$$

which is holomorphic in z . □

Corollary 4.11 *Suppose that $P(w) = w^n + a_1 w^{n-1} + \cdots + a_n$ (with a_i holomorphic functions defined on a neighborhood of z) has n distinct solutions $w_1, \dots, w_n$ at z . Then there exists unique holomorphic functions $f_1, \dots, f_n$ (defined on perhaps smaller neighborhood of z) with $f_i(z) = w_i$ satisfying $P(f_i) = 0$ so that $P(w) = \prod_1^n (w - f_i)$.*

4.1.1 Riemann-Hurwitz formula

Any compact Riemann surface can be described as a branched covering of $\mathbb{CP}^1$ once we admit the existence of at least one non-constant meromorphic function. From that description we can easily compute the genus of the surface. We state a more general version of that computation valid for a covering between compact surfaces.

Theorem 4.12 *Let $Y \rightarrow X$ be a branched covering of degree d between compact surfaces. For each branching point $y \in Y$, let $o(y)$ be its branching order. Then*

$$\chi(Y) = d\chi(X) - \sum (o(y) - 1).$$

Proof. The proof of the theorem follows from the existence of a triangulation with vertices containing the ramification locus, that is, the image of all ramification points by the covering map. We will assume the existence of that triangulation of X . If the simplices of this triangulation are sufficiently small, the inverse image of the triangulation is a triangulation of Y . The number of its simplices is d times the number of original simplices, except for the vertices. Each ramification point diminishes by $(o(y) - 1)$ the maximum number of d times the number of vertices of the original triangulation. □

4.1.2 Hyperelliptic Riemann surfaces

Let $f(z) = (z - a_1) \cdots (z - a_k) \in \mathcal{M}(\mathbb{CP}^1)$ with distinct roots $a_i \in \mathbb{C}$. The algebraic function defined by $P(z, w) = w^2 - f$ is a Riemann surface together with a branched covering of degree two which is branched on $a_1, \dots, a_k$ if k is even and on $a_1, \dots, a_k, \infty$ if k is odd. These Riemann surfaces are called hyperelliptic.

Observe that in that case the algebraic curve $\{ (z, w) \in \mathbb{C}^2 \mid P(z, w) = 0 \}$ is a Riemann surface by the implicit function theorem as at each solution (z, w) we have $P_z \neq 0$ or $P_w \neq 0$.

To understand the topology of hyperelliptic Riemann surfaces, consider the Riemann-Hurwitz formula to compute their genera. Let X_f be the Riemann surface as defined by $P(z, w) = w^2 - f$. If k is even we obtain

$$\chi(X_f) = 2\chi(\mathbb{CP}^1) - k = 4 - k$$

and as the Euler characteristic is given by $\chi = 2 - 2g$, we obtain $g = \frac{-2+k}{2} = k/2 - 1$. In the case k is odd we obtain

$$\chi(X_f) = 2\chi(\mathbb{CP}^1) - (k+1) = 3 - k$$

so that $g = \frac{-1+k}{2} = (k-1)/2$. In particular, for $k = 3$ we obtain an elliptic curve.

4.1.3 Exercises

1. Determine the Riemann surface defined by $P(z, w) = z^2 - w^3$ over $\mathbb{CP}^1$.
2. Determine the genus of the Riemann surface defined by $P(z, w) = z^n + w^n - 1$ over $\mathbb{CP}^1$.
3. The field $\mathcal{M}(\mathbb{CP}^1)$ is $\mathbb{C}(z)$, a purely transcendental extension of $\mathbb{C}$.
4. The field of meromorphic functions of a compact Riemann surface X is finitely generated over $\mathbb{C}$ and of transcendence degree one. (Use the fact that there exists a meromorphic function defined on X).
5. Any finitely generated field of transcendence degree one over $\mathbb{C}$ is isomorphic to the field of meromorphic functions of a compact Riemann surface.

4.1.4 Belyi's theorem

As an application of the construction of a Riemann surface of an algebraic function we will describe a relation between the number of branching points of the covering and the field of definition of an algebraic function.

We say that the Riemann surface X is defined over $\bar{\mathbb{Q}}$ if it is constructed as above starting with an irreducible polynomial in $\bar{\mathbb{Q}}[z, w]$, where $\bar{\mathbb{Q}}$ is the field of algebraic numbers.

Theorem 4.13 (Belyi) *A compact Riemann surface X is defined over $\bar{\mathbb{Q}}$ if and only if there exists a holomorphic covering $\pi : X \rightarrow \mathbb{CP}^1$ branched on three points.*

Proof. We will prove the “only if” part. The other implication being outside our scope because it needs basic algebraic geometry. We start with a polynomial $P \in \bar{\mathbb{Q}}[z, w]$. By theorem 4.6 there exists $\varphi : X \rightarrow \mathbb{CP}^1$ which is branched over a finite set S of algebraic points. We divide the proof in two steps:

1. We first modify this branched covering to a covering which is branched over rational points. Take $s \in S$ and let $h \in \mathbb{Q}[X]$ be its minimal polynomial. The map $h \circ \varphi : X \rightarrow \mathbb{CP}^1$ is a branched covering with branching points contained in $h(S) \cup \{h(z) \mid h'(z) = 0\}$. Observe that $h(s) = 0$ so we made one of the branching points in S rational at the cost of introducing new branching points. But the minimal polynomial of a point $z_0 \in \{z \mid h'(z) = 0\}$ is of degree strictly smaller than the degree of h and therefore the minimal polynomial of $h(z_0) \in \{h(z) \mid h'(z) = 0\}$ has strictly smaller degree too (being in the same

field extension as $\mathbb{Q}(z_0)$). We repeat this procedure with each element in S and obtain, by composing with each minimal polynomial, a branched covering where the new branching points have minimal polynomials of strictly smaller degrees. Eventually the degree is one and we obtain only rational branching points.

2. By the previous step, we may suppose that $\varphi : X \rightarrow \mathbb{C}P^1$ is branched on rational points. Now we reduce the number of branching points to at most three. Supposing it is greater than three, we can always assume that $\{0, 1, \infty\}$ are among those points by composing with an automorphism of $\mathbb{C}P^1$. For $m, n \in \mathbb{Z}^*$ such that $m + n \neq 0$, consider the map $f_{mn} : \mathbb{C}P^1 \rightarrow \mathbb{C}P^1$ defined by

$$f_{mn}(z) = \frac{(m+n)^{m+n}}{m^n n^n} z^m (1-z)^n.$$

The critical values are computed solving $f'_{mn}(z) = 0$ and we obtain that they are contained in $\{0, 1, \infty, \frac{m}{m+n}\}$. But the branching points are contained in $\{0, 1, \infty\}$. We conclude that for each rational branching point of φ outside $\{0, 1, \infty\}$ we can find a map f_{mn} so that $f_{mn} \circ \varphi$ transforms this branching point to one of $\{0, 1, \infty\}$. This concludes the proof.

□

5 Riemann surfaces as quotients.

One of the problems concerning Riemann surfaces is their classification. A natural classification is up to equivalence under biholomorphisms.

We first define maps between complex manifolds.

Definition 5.1 *Let M, N be two complex manifolds. A continuous map $F : M \rightarrow N$ is said to be a holomorphic map if for all charts $z_\alpha : U_\alpha \rightarrow \mathbb{C}^n$ and $w_\beta : W_\beta \rightarrow \mathbb{C}^n$ of M and N respectively,*

$$w_\beta \circ F \circ z_\alpha^{-1} : z_\alpha(U_\alpha) \rightarrow w_\beta(W_\beta)$$

is holomorphic.

A homeomorphism between complex manifolds which is holomorphic and whose inverse is also holomorphic is called a biholomorphism. The group of biholomorphisms of a complex manifold M is also called the automorphism group of M .

The complex projective space $\mathbb{C}P^n$ is the quotient of $\mathbb{C}^{n+1} - 0$ by the $\mathbb{C}^*$ -action $\lambda(z_1, \dots, z_{n+1}) = (\lambda z_1, \dots, \lambda z_{n+1})$. The orbit containing the point $(z_1, \dots, z_{n+1})$ is denoted $[z_1, \dots, z_{n+1}]$ (the homogeneous coordinates). Natural charts are given by defining the open sets $U_i = \{ [z_1, \dots, z_{n+1}] \mid z_i \neq 0 \}$ and $\varphi_i : U_i \rightarrow \mathbb{C}^n$ as

$$\varphi_i([z_1, \dots, z_{n+1}]) = \left(\frac{z_1}{z_i}, \dots, 1, \dots, \frac{z_{n+1}}{z_i} \right)$$

where the coordinate 1 corresponds to z_i/z_i should be deleted in the identification with $\mathbb{C}^n$. The transition functions are given by

$$\varphi_j \circ \varphi_i^{-1}(w_1, \dots, w_{n+1}) = \left(\frac{w_1}{w_j}, \dots, \frac{w_{n+1}}{w_j} \right)$$

where we think $(w_1, \dots, w_{n+1})$ as having the i -coordinate equal to 1 and $(\frac{w_1}{w_j}, \dots, \frac{w_{n+1}}{w_j})$ having the j -coordinate equal to 1.

Example 5.2 *The map $z \rightarrow (\mathcal{P}(z), \mathcal{P}'(z), 1)$ for $z \in \mathbb{C} - \Gamma_\tau$ and $z \rightarrow (0, 1, 0)$ for $z \in \Gamma_\tau$ defines a holomorphic embedding $E_\tau \rightarrow \mathbb{C}P^2$ whose image is the algebraic curve*

$$y^2 z = 4x^3 - g_2(\tau)xz^2 - g_3(\tau)z^3.$$

Example 5.3 *The disc and the half plane are biholomorphic complex manifolds of dimension 1. In fact, the map $z \rightarrow \frac{z-i}{z+i}$ is a biholomorphism from the half plane to the disc.*

Example 5.4 *The Möbius transformations preserving the disc are of the form*

$$z \rightarrow e^{i\theta} \frac{z + a}{\bar{a}z + 1}$$

.

The most fundamental theorem concerning complex manifolds of dimension one is Riemann's mapping theorem. We will state it without proof.

Theorem 5.5 (Riemann mapping theorem) *A simply connected one dimensional manifold is biholomorphic to either*

1. $\mathbb{C}P^1$ (the Riemann sphere)
2. $\mathbb{C}$
3. $H_{\mathbb{C}}^1 = \{z \in \mathbb{C}, |z| < 1\}$.

5.1 Automorphism groups

It will be important to determine for each manifold M its group of biholomorphisms $\text{Aut}(M)$. In the following theorem we need to recall Schwarz lemma:

Lemma 5.6 *If $f : H_{\mathbb{C}}^1 \rightarrow H_{\mathbb{C}}^1$ be a holomorphic map and $f(0) = 0$ then $|f(z)| \leq |z|$ for all $z \in H_{\mathbb{C}}^1$ and $|f'(0)| \leq 1$. If $|f'(0)| = 1$ or if $f(z) = z$ for some $z \neq 0$ then $f(z) = e^{i\theta}z$.*

Theorem 5.7 *The automorphism groups of the simply connected Riemann surfaces are*

1. $\text{Aut}(\mathbb{C}P^1) = \text{PSL}(2, \mathbb{C}) = \text{SL}(2, \mathbb{C})/\{\pm I\}$, all Möbius transformations

2. $Aut(\mathbb{C}) = \{az + b \mid a \neq 0, b \in \mathbb{C}\}$

3. $Aut(H_{\mathbb{C}}^1) = PSU(1,1) = SU(1,1)/\{\pm I\}$, Möbius transformations preserving the disc.

Proof. We first describe $f \in Aut(\mathbb{C})$. We have $f(z) = a_0 + a_1z + \dots$. As f is an automorphism, the image of a neighborhood of infinity is a neighborhood of infinity. Therefore it can be extended to a holomorphic function at infinity. Therefore $f(z)$ is a polynomial and by the fundamental theorem of algebra, it must be linear.

To show 1. observe that we can write in homogeneous coordinates $\mathbb{CP}^1 = \{[z_0, z_1]\}$, where z_0, z_1 are not both null. Any transformation of the form $[z_0, z_1] \rightarrow [az_0 + bz_1, cz_0 + dz_1]$, with $ad - bc \neq 0$ is an automorphism. So we have an action $PSL(2, \mathbb{C}) \times \mathbb{CP}^1 \rightarrow \mathbb{CP}^1$. Given an element $\gamma \in Aut(\mathbb{CP}^1)$ we can find an element $\gamma_1 \in PSL(2, \mathbb{C})$ such that $\gamma \circ \gamma_1(\infty) = \infty$. So $\gamma \circ \gamma_1 \in Aut(\mathbb{C})$ and we conclude using the description of $Aut(\mathbb{C})$.

To show 3. we observe first that $PSU(1,1) \subset PSL(2, \mathbb{C})$. That is, $SU(1,1) = \{A \in SL(2, \mathbb{C}) \mid h(Az, Az) = h(z, z)\}$, where $h(z, w) = z_0\bar{w}_0 - z_1\bar{w}_1$ is a hermitian form. So $PSU(1,1)$ preserves the disc $H_{\mathbb{C}}^1 = \{z \in \mathbb{CP}^1 \mid h(z, z) < 0\}$. If $\gamma \in Aut(H_{\mathbb{C}}^1)$, there exists an element $\gamma_1 \in PSU(1,1)$ such that $\gamma \circ \gamma_1(0) = 0$. By Schwarz's lemma we obtain $|f'(0)| \leq 1$ and, as f is a biholomorphism, the same inequality for the inverse function gives $|f'(0)| = 1$. By Schwarz's lemma we conclude that $\gamma \circ \gamma_1(z) = e^{i\theta}z$ and that concludes the proof. $\square$

Corollary 5.8 *A Riemann surface covered by $\mathbb{CP}^1$ is biholomorphic to $\mathbb{CP}^1$.*

Proof. This follows from the fact that any Möbius transformation has a fixed point. It implies that there is no subgroup of the Möbius group acting freely on $\mathbb{CP}^1$. $\square$

On the other hand observe that the involution $\iota : z \rightarrow -\frac{1}{\bar{z}}$ defined on $\mathbb{CP}^1$ does not have fixed points. The quotient space $\mathbb{CP}^1/\langle \iota \rangle$ is the real projective plane which is not a Riemann surface.

Exercise 5.9 *A meromorphic function on $\mathbb{CP}^1$ is a holomorphic map of $\mathbb{CP}^1$ on itself. They are all rational functions, that is $f(z) = \frac{p(z)}{q(z)}$ where $p(z)$ and $q(z)$ are polynomials.*

Exercise 5.10 *If K is a field $PSL(n, K) = PGL(n, K)$ if and only if every element of K has an n -th root. For instance $PSL(2, \mathbb{R}) \neq PGL(2, \mathbb{R})$.*

Exercise 5.11 *$PU(1,1)$ acts doubly transitively on the boundary. That is given $x_1, y_1, x_2, y_2 \in \partial H^1\mathbb{C}$ with $x_i \neq y_i$, there exists an element $\gamma \in PU(1,1)$ such that $\gamma x_1 = x_2$ and $\gamma y_1 = y_2$.*

5.1.1 Conjugacy classes

It is very important to understand the conjugacy classes of elements in the automorphism groups. Elements in the same conjugacy class act in an “equivalent” way.

Lemma 5.12 *An element of $PSL(2, \mathbb{C})$ has one or two fixed points. We have*

1. *If it has only one fixed point then it is conjugate to $z \rightarrow z + 1$.*
2. *If it has two fixed points it is conjugate to $z \rightarrow \lambda z$, $\lambda \neq 1, 0$.*

Proof. Given any Möbius transformation we solve the equation

$$\frac{az + b}{cz + d} = z.$$

It has one or two solutions. If it has only one solution by conjugating with an element of $PSL(2, \mathbb{C})$ we can suppose that ∞ is that fixed point. In that case the element must be of the form $z \rightarrow az + b$. We immediately see that $a = 1$ otherwise there would be a second fixed point. Moreover, by conjugating with $z \rightarrow \frac{1}{b}z$ we obtain $z \rightarrow z + 1$. To show the second part we observe that we can conjugate an element with two fixed points to one fixing 0 and ∞ . That gives clearly the form $z \rightarrow \lambda z$. $\square$

We can further refine that lemma to obtain the orbit space by the conjugation action of $PSL(2, \mathbb{C})$. The proof of the following theorem is a simple consequence of the lemma.

Theorem 5.13 *The conjugacy classes of $PSL(2, \mathbb{C})$ are uniquely represented by the following elements*

1. *$z \rightarrow z + 1$ called parabolic.*
2. *$z \rightarrow e^{i\theta}z$, $0 \leq \theta \leq \pi$, called elliptic.*
3. *$z \rightarrow \lambda z$, $\lambda \in \mathbb{C}$ $|\lambda| > 1$, called loxodromic. In the case $\lambda \in \mathbb{R}$ we call it a hyperbolic transformation.*

Proof. The first part is contained in the previous lemma. For the second and third part we observe that if $\gamma(z) = \lambda z$, in order to preserve the fixed points, we are allowed to conjugate by elements of the form $z \rightarrow az$, which commute with γ (so irrelevant), or $z \rightarrow a/z$. In that case γ is transformed to $g\gamma g^{-1}(z) = \frac{1}{\lambda}z$. This shows the result. $\square$

Considering only elements in $PSU(1, 1)$ we describe conjugacy classes in the following definition.

Definition 5.14 *$\gamma \in PSU(1, 1)$ is called*

1. *Elliptic if it has a fixed point in $H_{\mathbb{C}}^1$.*

2. Parabolic if it has a unique fixed point in $\partial H_{\mathbb{C}}^1$.

3. Hyperbolic if it has two fixed points in $\partial H_{\mathbb{C}}^1$.

Theorem 5.15 *Let $\gamma \in PSU(1, 1)$ and consider a lift $\tilde{\gamma} \in SU(1, 1)$. Then γ is*

1. elliptic if and only if $\text{tr}^2 \tilde{\gamma} < 4$,
2. parabolic if and only if $\text{tr}^2 \tilde{\gamma} = 4$ and γ is not the identity,
3. hyperbolic if and only if $\text{tr}^2 \tilde{\gamma} > 4$.

Observe, however, that conjugation in $PSU(1, 1)$ splits certain conjugacy classes in $PSL(2, \mathbb{C})$ (of course, some disappear). For instance, the parabolic class is split in two: $z \rightarrow z + 1$ and $z \rightarrow z - 1$. Analogously, the elliptic class $z \rightarrow e^{i\theta}$, $0 \leq \theta \leq \pi$ splits in two, so that $0 \leq \theta < 2\pi$ is the parameterization of the classes. On the other hand, the only loxodromic classes which appear in $PSL(2, \mathbb{C})$ are those with $\lambda > 1$ and they don't split.

Remark 5.16 *Let $\widehat{PSU}(1, 1) = \langle PSU(1, 1), z \rightarrow \bar{z} \rangle$. Using conjugation on that group we can collapse again the splitting. In particular $z \rightarrow z + 1$ and $z \rightarrow z - 1$ are conjugate in the corresponding group $\widehat{PSL}(2, \mathbb{R})$.*

5.2 The complex plane $\mathbb{C}$ and its quotients

Theorem 5.17 *A Riemann surface is covered by $\mathbb{C}$ if and only if it is biholomorphic to $\mathbb{C}$, $\mathbb{C} - 0$ or a torus.*

Proof. We prove the only if part. The other implication is a consequence of the next proposition. Let $\Gamma \subset Aut(\mathbb{C})$ be the covering group, where $Aut(\mathbb{C})$ is the group of biholomorphisms of $\mathbb{C}$. If $\gamma(z) = az + b$ is a non trivial element of Γ then $a = 1$, otherwise γ would have a fixed point. So Γ is generated by translations. We saw in theorem 3.21 that a discrete subgroup of $Aut(\mathbb{C})$ generated by translations is one of the following:

1. $\{id\}$
2. $\langle \gamma \rangle = \mathbb{Z}$, a group generated by one translation $\gamma(z) = z + \omega$
3. $\langle \gamma_1, \gamma_2 \rangle = \mathbb{Z} \oplus \mathbb{Z}$, a group generated by two translation $\gamma_1(z) = z + \omega_1$ and $\gamma_2(z) = z + \omega_2$ with ω_1 and ω_2 linearly independent over $\mathbb{R}$.

The first case corresponds to $\mathbb{C}$. For the second case the function $z \rightarrow e^{2\pi z/\omega}$ establishes a biholomorphism between $\mathbb{C}/\langle \gamma \rangle$ and $\mathbb{C} - 0$. In the third case the quotient manifold is diffeomorphic to a torus.

□

To complete the theorem we need to show that any torus is covered by $\mathbb{C}$. That is, the complex disc cannot cover a torus. This follows from the following proposition.

Proposition 5.18 *Let $\Gamma \subset \text{Aut}(H_{\mathbb{C}}^1)$ be a discrete group without fixed points. If Γ is abelian, then it is cyclic.*

Proof. We separate in two cases. If $\gamma \in \Gamma$ is parabolic we can, without loss of generality, suppose that $\gamma(z) = z + a$. A computation then shows that any commuting element is parabolic. In fact

$$\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$$

implies

$$\begin{pmatrix} a + xc & b + xd \\ c & d \end{pmatrix} \begin{pmatrix} a & ax + b \\ c & cx + d \end{pmatrix}$$

So $xc = 0$ and $x(a - d) = 0$ which implies $c = 0$ and $a = d$. That is, the commuting element is parabolic. By discreteness we obtain that it is cyclic.

Analogously, if γ is hyperbolic, without loss of generality, suppose that $\gamma(z) = \lambda z$. We easily conclude (by the lemma below) that an element commuting with it is of the same form and using discreteness we conclude that the subgroup is cyclic. $\square$

Lemma 5.19 *Two hyperbolic elements commute if and only if they have the same fixed points.*

Proof. We write one element as $z \rightarrow \lambda z$ and the other by a general Möbius transformation. Then, by commutativity

$$\begin{pmatrix} \lambda^{-1/2} & 0 \\ 0 & \lambda^{1/2} \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \lambda^{1/2} & 0 \\ 0 & \lambda^{-1/2} \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

A computation shows that $b = c = 0$. $\square$

A proof without computations runs as follows. Observe that if G (any group) acts on M (any space) and g_1 commutes with g_2 the fixed points of g_1 are preserved by g_2 and the fixed points of g_2 are preserved by g_1 , in fact

$$g_1(x) = x \rightarrow g_2 g_1(x) = g_2(x) \rightarrow g_1(g_2(x)) = g_2(x).$$

The proposition is a refinement of the observation: If γ has only one fixed point any commuting element will have precisely the same fixed point (so if γ is parabolic the commuting element is also parabolic). If γ has two fixed points the commuting element either has the same fixed points or it interchanges the two fixed points. But in the last case, if $\gamma \in \text{Aut}(H_{\mathbb{C}}^1)$ then the commuting element must have only one fixed point (in fact it is elliptic) and therefore γ has only one fixed point, a contradiction.

5.3 Fuchsian groups

Definition 5.20 *A Fuchsian group is a discrete subgroup of $PSU(1, 1)$.*

In order to define a quotient of the disc by a discrete group as a Riemann surface we need to verify that the action is free and properly discontinuous. The action is free if there are no elliptic elements, also called torsion elements. On the other hand, the action is always properly discontinuous as is shown by the next theorem.

Theorem 5.21 *A subgroup $\Gamma \subset Aut(H_{\mathbb{C}}^1)$ is Fuchsian if and only if it acts properly discontinuously.*

Proof. Clearly if Γ acts properly discontinuously then it is discrete. Now suppose it is discrete and it does not act properly discontinuously.

Recall the normal family theorem:

Theorem 5.22 (Normal family theorem) *Suppose $f_n : \Omega \rightarrow \mathbb{C}$ is a family of holomorphic functions defined on a region of $\mathbb{C}$. If f_n is uniformly bounded on each compact subset of Ω (a normal family) then there exists a subsequence which converges uniformly on compact subsets (the limit function will then be holomorphic)*

We need the following lemma

Lemma 5.23 *If a sequence $\gamma_n \in Aut(H_{\mathbb{C}}^1)$ converges uniformly on compact subsets to γ then*

1. $\gamma \in Aut(H_{\mathbb{C}}^1)$ or
2. γ is a constant function with value some $e^{i\theta}$.

Proof. If $\gamma_n(x_0) \rightarrow b$ with $|b| = 1$ then by the maximum modulus principle $\gamma(x_0) = b = \gamma(z)$, for all $z \in H_{\mathbb{C}}^1$. Otherwise we have $\gamma : H_{\mathbb{C}}^1 \rightarrow H_{\mathbb{C}}^1$ and taking a subsequence γ_n^{-1} converges uniformly on compact subsets to γ^{-1} so $\gamma \in Aut(H_{\mathbb{C}}^1)$. □

Back to the proof: if the action is not properly discontinuous there exists a compact $K \subset H_{\mathbb{C}}^1$ and a sequence of distinct elements $\gamma_n \in \Gamma$ such that $\gamma_n(K) \cap K \neq \emptyset$. Clearly the sequence γ_n is a normal family. Therefore, taking perhaps a subsequence, it converges uniformly on compact subsets to a holomorphic function. Taking a subsequence if necessary we have $\gamma_n(x_n) = y_n$ with $\lim x_n = x$ and $\lim y_n = y$, therefore $\lim \gamma_n(x) = y$. We conclude, using the lemma, that γ_n converges to an element of $Aut(H_{\mathbb{C}}^1)$, therefore the group is not discrete. □

The following lemma is an important technical component of the next theorem.

Lemma 5.24 (Shimizu) *If $z \rightarrow z + 1$ belongs to a Fuchsian group in $PSL(2, \mathbb{R})$, then every other element γ of the form*

$$\frac{az + b}{cz + d}$$

satisfies $|c| \geq 1$, provided $c \neq 0$.

Proof. We set

$$A_1 = \begin{pmatrix} a & b \\ c & d \end{pmatrix} A_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and define by induction for $n \geq 1$,

$$A_{n+1} = A_n A_0 A_n^{-1}.$$

We compute the coefficients of A_{n+1} obtaining

$$\begin{aligned} a_{n+1} &= 1 - c_n a_n \\ b_{n+1} &= a_n^2 \\ c_{n+1} &= -c_n^2 \\ d_{n+1} &= 1 + a_n c_n \end{aligned}$$

If $c < 1$ then c_n converges, in fact $|c_n| = |c|^{2^{n-1}}$. We claim that $\lim a_n = 1$. Observe that $|a_{n+1}| \leq 1 + |a_n c_n| \leq 1 + |a_n|$. By induction then $|a_{n+1}| \leq n + |a|$. We obtain then $|a_{n+1}| \leq 1 + |a_n c_n| \leq 1 + |c_n|(n + |a|) \leq 1 + |c|^{2^{n-1}}(n + |a|)$ and the result follows. $\square$

A Fuchsian group $\Gamma \subset PSL(2, \mathbb{R})$ is said to be co-compact if the quotient $H_{\mathbb{C}}^1/\Gamma$ compact. From Shimizu lemma we conclude the following theorem which says that if a Riemann surface is compact and not the sphere or a quotient of the complex plane then its fundamental group does not have parabolics.

Theorem 5.25 *If $\Gamma \subset PSL(2, \mathbb{R})$ is co-compact without torsion then any non-trivial element is hyperbolic.*

Proof. If there were a parabolic element, by conjugation we may suppose it $z \rightarrow z + 1$ and generator of the parabolic group Γ_{∞} fixing ∞ . As

$$Im(\gamma(z)) = \frac{Im(z)}{|cz + d|^2}$$

for any $\gamma(z) = \frac{az+b}{cz+d}$ in Γ we estimate using Shimizu's lemma that if $Im(z) > 1$ then

$$Im(\gamma(z)) \leq \frac{1}{|c|^2 Im(z)} < 1$$

for γ not in Γ_{∞} . Therefore the set $\{ z \mid -\frac{1}{2} < \operatorname{Re} z < \frac{1}{2}, Im(z) > 1 \}$ passes to the quotient, but it is not compact, a contradiction. $\square$

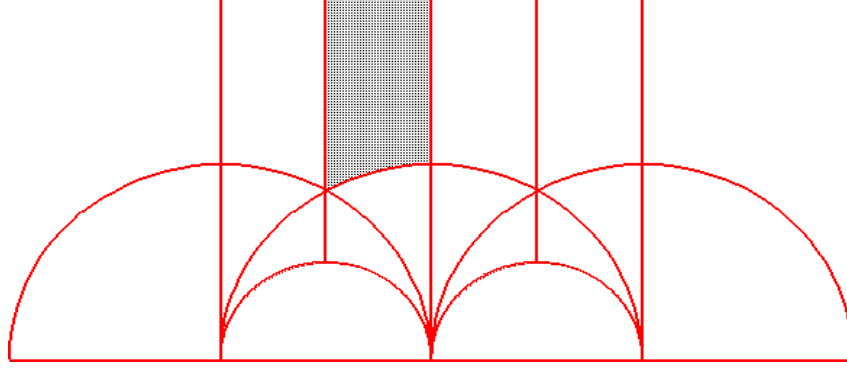


Figure 4: A fundamental domain for a triangle group containing $PSL(2, \mathbb{Z})$ as an index two subgroup. The fundamental domain for $PSL(2, \mathbb{Z})$ is the symmetric double of the grey region.

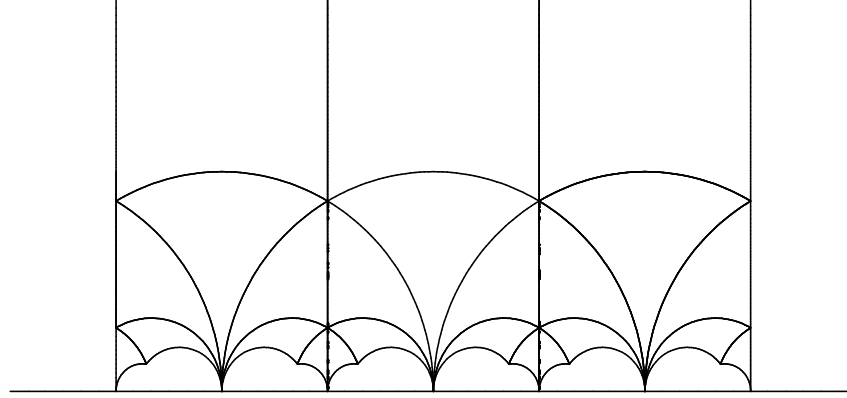


Figure 5: A fundamental domain for $PSL(2, \mathbb{Z})$ and some of its translates.

5.4 Fundamental domains

Definition 5.26 A fundamental domain of a properly discontinuous action on a topological manifold, $\Gamma \times X \rightarrow X$ is an open set $F \subset X$ such that

1. $\bigcup_{\gamma \in \Gamma} \gamma \overline{F} = X$, where $\overline{F}$ is the closure of F
2. If $x, y \in F$ they are not in the same orbit.

We do not suppose that the action is free but observe that a fixed point of an element in Γ is never contained in F . It might be contained in the closure of F .

Example 5.27 A fundamental domain for the action of the additive group generated by the translations $z \rightarrow z + 1$ and $z \rightarrow z + \tau$ is the parallelogram defined by the sides $1, \tau$.

5.4.1 $PSL(2, \mathbb{Z})$

Theorem 5.28 $D = \{ z \in H_{\mathbb{C}}^1 \mid |z| > 1, -1/2 < \text{Re}(z) < 1/2 \}$ is a fundamental domain for $PSL(2, \mathbb{Z})$.

Proof. Again we use

$$Im(\gamma(z)) = \frac{Im(z)}{|cz + d|^2}$$

to observe that fixing $\tau \in \mathbb{C}$, there is only a finite number of elements $\gamma \in PSL(2, \mathbb{Z})$ with $|c\tau + d|^2 < M$ for a fixed bound M . This follows because $\mathbb{Z}\tau + \mathbb{Z}$ is a discrete group. Take γ such that $Im(\gamma(\tau))$ is maximum. Using the translation we can suppose without loss of generality that $-1/2 \leq Re(\tau) \leq 1/2$. We claim that $|\gamma(\tau)| \geq 1$, otherwise using the inversion $s(z) = -1/z$ we would get $Im(s\gamma(\tau)) = \frac{Im(\gamma(\tau))}{|\gamma(\tau)|^2} > Im(\gamma(\tau))$. A contradiction.

Suppose now that τ and $\gamma(\tau)$ belong to $\bar{D}$. Without loss of generality we may assume that $Im(\gamma(\tau)) \geq Im(\tau)$. Therefore

$$|c\tau + d| \leq 1.$$

Just looking at the imaginary part, that is, $Im(c\tau + d) = cIm\tau \geq c\frac{\sqrt{3}}{2}$, we obtain that the only possibilities are $c = 0, 1, -1$. If $c = 0$ it follows easily that γ is either the translation or the identity. If $c = 1$, we must have $|z + d| \leq 1$. We claim that that is only possible if $z = \omega$ or $z = -\bar{\omega}$ or $z = i$. That can be seen easily in the picture. Analogously we obtain those two points if $c = -1$. □

5.5 $\Gamma(2)$

Let $\pi_N : SL(2, \mathbb{Z}) \rightarrow SL(2, \mathbb{Z}_N)$ be the homomorphism obtained by reducing modulo N . It passes to the quotients

$$\varphi_N : SL(2, \mathbb{Z})/\{I, -I\} \rightarrow SL(2, \mathbb{Z}_N)/\{I, -I\}.$$

The kernel of this homomorphism is called the principal congruence group of level N , $\Gamma(N) \subset PSL(2, \mathbb{Z})$.

The simplest case, $\Gamma(2)$, acts freely on the complex disc so that $H_{\mathbb{C}}^1/\Gamma(2)$ is a sphere with three points deleted.

To understand the action, observe first that the homomorphism φ_N is clearly surjective and, as $SL(2, \mathbb{Z}_2) = PSL(2, \mathbb{Z}_2)$ has 6 elements which can easily be enumerated:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

we have, therefore, that $\Gamma(2) \subset PSL(2, \mathbb{Z})$ is of index 6.

The fundamental domain of subgroups of finite index can be computed using the following lemma.

Lemma 5.29 *Suppose D is a fundamental domain for a group G acting on a space M . Let $H \subset G$ be a subgroup of index k and $Hg_1, \dots, Hg_k$ be its left cosets. Then $D_H = g_1D \cup \dots \cup g_kD$ is a fundamental domain for H .*

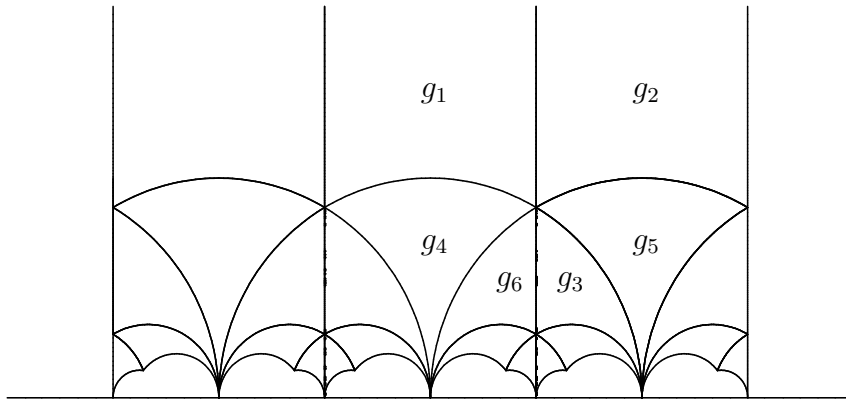


Figure 6: A fundamental domain for $\Gamma(2)$ showing the six translates of the fundamental region of $PSL(2, \mathbb{Z})$ corresponding to each coset.

Proof. If $x, y \in D_H$ and there exists $h \in H$ such that $y = hx$ then, as $x \in g_i D$ and $y \in g_j D$, we might suppose that $g_j \bar{y} = h g_i \bar{x}$ for $\bar{x}, \bar{y} \in D$. That is, $\bar{y} = g_j^{-1} h g_i \bar{x}$ which contradicts the fact that D is a fundamental domain for G . On the other hand, $H\overline{D_H} = M$ follows because $G = \bigcup Hg_i$. $\square$

Left coset representatives of $\Gamma(2)$ are obtained by choosing an inverse image for each element of $SL(2, \mathbb{Z}_2)$:

$$g_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} g_2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} g_3 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$$g_4 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} g_5 = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} g_6 = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}.$$

The boundary of the fundamental domain consists of 2 vertical half lines paired by the parabolic element

$$\gamma_1 = z \rightarrow z + 2$$

and two pairs of arcs paired by parabolic elements in the group:

$$\gamma_2 = g_4 \gamma_1 g_4^{-1} = z \rightarrow \frac{z}{2z + 1}$$

for the sides of the region $g_4 D \cup g_6 D$ (where D is the fundamental domain for $PSL(2, \mathbb{Z})$ found before),

$$z \rightarrow \frac{3z - 2}{2z - 1}$$

for the sides of the region $g_3 D \cup g_5 D$. One should observe that the three points of $H_{\mathbb{C}}^1$ in the boundary of the region are identified by those pairings and, around that point, the regions match together to form a complex disc. The quotient is the sphere where 3 points are deleted.

6 Hyperbolic geometry

An important development was the discovery by Poincaré was that Möbius transformations preserving the disc were, in fact, isometries of the disc equipped with a metric of constant negative curvature.

6.1 Riemannian manifolds

A Riemannian manifold is a manifold equipped with a positive definite scalar product $\langle \cdot, \cdot \rangle$ defined on the tangent space at each point. Using the Riemannian metric one defines the length of curves and a metric on the manifold so that the distance between two points is the infimum of all lengths of curves joining them:

$$d(p, q) = \inf_{\gamma(0)=p, \gamma(1)=q} L(\gamma)$$

where

$$L(\gamma) = \int_0^1 \sqrt{\langle \dot{\gamma}, \dot{\gamma} \rangle} dt$$

The group of isometries, that is, distance preserving diffeomorphisms of a metric space M , will be denoted by $Isom(M)$. Isometries are determined by their derivative at one point:

Theorem 6.1 *Let M be a connected Riemannian manifold, $\varphi : M \rightarrow M$ be an isometry with $\varphi(p) = p$. Then $\varphi_* : T_p M \rightarrow T_p M$ determines φ .*

Exercise 6.2 *Let E^n be the n -dimensional Euclidean space. Show that $Isom(E^n)$ is the group $\{x \rightarrow Ax + B\}$ where A is orthogonal.*

Exercise 6.3 *Prove the following exact sequence*

$$0 \rightarrow \mathbb{R}^n \rightarrow Isom(E^n) \rightarrow O(n) \rightarrow 1$$

Exercise 6.4 *The finite subgroups of $O(2)$ are the cyclic group generated by a rotation and the dihedral group generated by two reflections.*

The discrete subgroups of $Isom(E^2)$ were classified in the 19th century. The classification starts writing the discrete group Γ inside the exact sequence

$$0 \rightarrow T \rightarrow \Gamma \rightarrow H \rightarrow 1$$

where T is the subgroup of translations of Γ and H is a subgroup of $O(2)$. As Γ is discrete, T is also discrete. Therefore it is either trivial or $\mathbb{Z}$ or $\mathbb{Z} \oplus \mathbb{Z}$. If T is trivial Γ is either finite cyclic or dihedral. If T has one generator it is one of the seven strip patterns. If T is a lattice it is one of the 17 crystallographic groups.

Example 6.5 The triangle groups are those groups generated by reflections in three lines. If the angles are π/p , π/q and π/r for positive integers p, q, r we should have $\pi/p + \pi/q + \pi/r = \pi$ and in this case the group is discrete. That gives three possibilities for (p, q, r) , that is, $(3, 3, 3)$, $(2, 3, 6)$ and $(2, 4, 4)$. The region inside the triangle is a fundamental domain for the triangle group. (reflections on the sides of the triangle of angles $2\pi/3$, $\pi/6$ and $\pi/6$ also defines a discrete group, this is the only non-obtuse triangle leading to a discrete group)

Example 6.6 The index two subgroup of orientation preserving isometries of a triangle group has two generators. If we denote r_1 , r_2 and r_3 the reflections on the sides of the triangles, the subgroup of orientation preserving isometries is generated by $r_1 \circ r_2$ and $r_1 \circ r_3$. A fundamental domain consists of any two adjacent triangles.

Exercise 6.7 1. Consider the Riemannian manifold obtained by identifying the two vertical lines $\{\operatorname{Re} z = 1\}$ and $\{\operatorname{Re} z = 2\}$ on the upper half-plane via the isometry $z \rightarrow z + 1$. Prove that this manifold is complete. Hint: show that any geodesic is defined on $\mathbb{R}$ by glueing copies of the vertical band to form the complete Poincaré half-plane.

2. Consider the Riemannian manifold obtained by identifying the vertical lines $\{\operatorname{Re} z = 1\}$ and $\{\operatorname{Re} z = 2\}$ on the upper half-plane via the isometry $z \rightarrow 2z$. That manifold is not complete. Prove that the sequence $(1, 2^i)$ is a Cauchy sequence but it is not convergent.

Local isometries between Riemannian spaces are very special:

Theorem 6.8 Let $d : M \rightarrow N$ be a surjective local isometry between Riemannian manifolds. If M is complete and connected then d is a covering.

6.2 Hyperbolic geometry

We will start with the half-plane model and define the metric

$$ds^2 = \frac{dx^2 + dy^2}{y^2} = \frac{|dz|^2}{\operatorname{Im}(z)^2}$$

Given a metric g on a Riemannian manifold we can define a volume form dv by imposing $dv(X_1, \dots, X_n) = 1$ for an orthonormal basis. In local coordinates we have $dv = \sqrt{\det(g_{ij})} dx^1 \cdots dx^n$. For hyperbolic geometry we get

$$dv = \frac{1}{y^2} dx dy.$$

Theorem 6.9 $PSL(2, \mathbb{R}) \subset \operatorname{Isom}(H_{\mathbb{C}}^1)$.

Proof. We need to show that

$$\frac{|d\gamma(z)|^2}{(\operatorname{Im}\gamma(z))^2} = \frac{|dz|^2}{(\operatorname{Im}z)^2}.$$

This follows from a simple computation. □

Theorem 6.10 *The geodesics of $H_{\mathbb{C}}^1$ are vertical lines or circles perpendicular to the $\mathbb{R}$ -axis.*

Proof. We first observe that given two points with the same x -coordinate, $p = (x, y_1)$ and $q = (x, y_2)$ (without loss of generality we suppose $y_2 > y_1$), then

$$d(p, q) = \inf \int \frac{\sqrt{dx^2 + dy^2}}{y}.$$

But $\int \frac{\sqrt{dy^2}}{y} \leq \int \frac{\sqrt{dx^2 + dy^2}}{y}$. As $\int \frac{\sqrt{dy^2}}{y} \geq \ln(y_2/y_1)$ we conclude that

$$d(p, q) = \ln(y_2/y_1).$$

We use now the fact that geodesics are preserved by isometries and that vertical lines are transformed to circles orthogonal to the real axis or to vertical lines by $PSL(2, \mathbb{R})$. $\square$

In the following we will call a hyperbolic triangle a simplex in $H_{\mathbb{C}}^1$ whose boundary is formed by three geodesic segments.

Theorem 6.11 *Let Δ be an hyperbolic triangle with angles α , β and γ . Then*

$$Area(\Delta) = \pi - \alpha - \beta - \gamma$$

Proof. Suppose first that the triangle has an ideal point, that is, one of the angles is null, or, equivalently, one of its vertices is in the boundary of $H_{\mathbb{C}}^1$. Without loss of generality we might suppose that the vertex is ∞ and one of the geodesics is the half circle of radius one centered at the origin. The other two are vertical lines which form angle α and β with the circle. then

$$Area(\Delta) = \int \int \frac{dx dy}{y^2} = \int_a^b dx \int_{\sqrt{1-x^2}}^{\infty} \frac{dy}{y^2} = \int_a^b \frac{dx}{\sqrt{1-x^2}}.$$

By a change of coordinate $x = \cos \theta$ we get

$$\int_{\pi-\alpha}^{\beta} \frac{-\sin \theta}{\sin \theta} d\theta = \pi - \alpha - \beta$$

If the triangle Δ_1 is compact we choose one of the vertices (say the one with angle γ) and prolong one of the sides containing it up to the boundary of $H_{\mathbb{C}}^1$. We have three triangles one (containing an ideal point) being the union of the other two. Comparing their areas:

$$\Delta_1 = \Delta_1 + \Delta_2 - \Delta_2$$

$$A(\Delta_1) = \pi - \alpha - (\beta - \theta) - (\pi - (\pi - \gamma) - \theta) = \pi - \alpha - \beta - \gamma.$$

$\square$

Decomposing a polygon in triangles we obtain the following

Corollary 6.12 *For a geodesic polygon with n sides denote by α the sum of the internal angles. Then*

$$A = n\pi - 2\pi - \alpha.$$

Using a geodesic triangulation one can prove Gauss-Bonnet theorem:

Theorem 6.13 *If S_g is a hyperbolic surface, then*

$$A = -2\pi\chi.$$

Proof. We have $A = \sum(\pi - \alpha_i - \beta_i - \gamma_i)$ summing over all triangles, say F of them. The angles sum to 2π times the number of vertices, say V . Therefore $A = \pi(F - 2V)$. On the other hand the number of edges is precisely $E = 3F/2$. We conclude that $\chi = F - E + V = F - 3F/2 + V = -F/2 + V = A/(-2\pi)$ $\square$

The Poincaré metric on the disc is given by

$$ds^2 = \frac{4|dz|^2}{(1 - |z|^2)^2}.$$

The geodesics of the hyperbolic disc are described in the following

Proposition 6.14 *The geodesics of the hyperbolic disc are sub-arcs of circles orthogonal to the boundary of the disc.*

Proposition 6.15 *Let S be a Riemann surface with a Fuchsian model $H_{\mathbb{C}}^1/\Gamma$. If γ is a hyperbolic element and L_γ the geodesic obtained by projection of its axis then*

$$|tr(\gamma)| = 2 \cosh \left(\frac{l(L_\gamma)}{2} \right)$$

where $l(L_\gamma)$ is the length of the geodesic.

Proof. We may assume that $\gamma(z) = \lambda^2 z$. Then $l(L_\gamma) = \int_1^{\lambda^2} \frac{dy}{y} = \ln \lambda^2$. $\square$

7 Teichmüller space and Moduli space

From the theory of covering spaces we obtain the following

Proposition 7.1 *Let $f : M \rightarrow M'$ be a continuous function. Then there exists a continuous function $\tilde{f} : \tilde{M} \rightarrow \tilde{M}'$, lift of f to the universal covers. It is unique up to a choice of an origin on a fiber of $\tilde{M}$ and a corresponding point on the fiber of $\tilde{M}'$.*

Clearly, if f is a homeomorphism the lift will be a homeomorphism. Also, if the manifolds are Riemannian and f is an isometry, the lift will be an isometry. Suppose now that $M = X/\Gamma$ and $M' = X/\Gamma'$ are Riemannian. Then a lift $\tilde{f}$ is an isometry $g : X \rightarrow X$ which is “equivariant” with respect to the action of Γ and Γ' . That is, by

uniquity of the lift and transitivity of the group of covering transformations, for any $\gamma \in \Gamma$ there exists $\gamma' \in \Gamma'$ such that

$$g\gamma x = \gamma'gx$$

valid for every $x \in X$, we conclude that

$$\Gamma' = g\Gamma g^{-1}.$$

We proved the following

Proposition 7.2 *If $M = X/\Gamma$ and $M' = X/\Gamma'$ are isometric then $\Gamma' = g\Gamma g^{-1}$.*

Of course, if M and M' are homeomorphic we conclude that $\Gamma' = f \circ \Gamma \circ f^{-1}$, where $f : X \rightarrow X$ is a homeomorphism. We will suppose $X = G/K$ is a homogeneous space with compact isotropy K . That is always the case if X is Riemannian with a transitive group of isometries.

7.1 The representation space $Hom(\Gamma, G)$

Assume Γ is finitely generated and G a Lie group. Two actions on $Hom(\Gamma, G)$ are essential

$$Aut(G) \times Hom(\Gamma, G) \rightarrow Hom(\Gamma, G)$$

given by

$$(\varphi, \rho) \rightarrow \varphi \circ \rho$$

and

$$Aut(\Gamma) \times Hom(\Gamma, G) \rightarrow Hom(\Gamma, G)$$

given by

$$(\varphi, \rho) \rightarrow \rho \circ \varphi^{-1}.$$

Definition 7.3 *The Teichmüller space $\mathbf{T}(\Gamma)$ in G is the set $Hom(\Gamma, G)/Aut(G)$.*

Remark: Sometimes one defines $R(\Gamma, G) = Hom(\Gamma, G)/G^0$, where G^0 is the connected component of G . If G is an algebraic group, a choice of presentation of Γ leads to a description of $Hom(\Gamma, G)$ as an algebraic variety. In order to obtain a structure of algebraic variety we consider the Mumford quotient which is written $\mathbb{R}(\Gamma, G) = Hom(\Gamma, G)/G^0$. This is called the *character variety*. The idea is to consider the subring of the ring defining $Hom(\Gamma, G)$ which consist of invariant functions.

We consider the action

$$Aut(\Gamma) \times \mathbf{T}(\Gamma) \rightarrow \mathbf{T}(\Gamma)$$

and observe that it is trivial when restricted to the inner automorphisms of Γ . We have therefore the action

$$Out(\Gamma) \times \mathbf{T}(\Gamma) \rightarrow \mathbf{T}(\Gamma)$$

where $Out(\Gamma) = Aut(\Gamma)/Inn(\Gamma)$.

Definition 7.4 *The Moduli space $\mathbf{M}(\Gamma)$ in G is the set $\mathbf{T}(\Gamma)/\text{Out}(\Gamma)$.*

We will also define $R(\Gamma, G) \subset \text{Hom}(\Gamma, G)$ as those representations which are faithful and discrete.

An equivalent description of the moduli space relating to the uniformization problem runs as follows

Definition 7.5 *The moduli space $\mathbf{M}$ of Γ in G is the set $\text{Hom}(\Gamma, G)/\equiv$, where we identify two representations if the image groups are conjugate.*

As observed above, a practical description is obtained if we mark generators for Γ . Suppose $\langle \gamma_i \rangle$ is a family of such generators. A representation is determined by the values $\pi(\gamma_i)$.

Definition 7.6 *The Teichmüller space $\mathbf{T}$ of the marked Γ in G is the set $\text{Hom}(\Gamma, G)/\equiv$, where we identify two representations if $\pi(\gamma_i) = g\pi'(\gamma_i)g^{-1}$.*

Clearly there is a canonical projection $\mathbf{T} \rightarrow \mathbf{M}$. The problem is that even if $\pi'(\Gamma) = g\pi(\Gamma)g^{-1}$ globally, it might not be true that $\pi'(\gamma_i) = g\pi(\gamma_i)g^{-1}$. It could happen that $g\pi(\gamma_i)g^{-1} = \pi'(\beta_i) \neq \pi'(\gamma_i)$. But if that is the case β_i would be new generators of Γ satisfying the same relations. This is possible if an element $\varphi \in \text{Aut}(\Gamma)$ satisfies $\varphi(\gamma_i) = \beta_i$.

7.2 Tori

7.2.1 Teichmüller space

A complex torus is covered by the complex plane $\mathbb{C}$. The Teichmüller space of representations of $\Gamma = \mathbb{Z} \oplus \mathbb{Z}$ in $\text{Aut}(\mathbb{C})$ is obtained by identifying generators (ω_1, ω_2) and (ω'_1, ω'_2) when they differ by a conjugation with an element of $\widehat{\text{Aut}}(\mathbb{C})$ (those are the automorphisms of the group of automorphisms of $\mathbb{C}$). That is $\omega'_i = \lambda\omega_i$ or $\omega'_i = \bar{\omega}_i$. We can therefore find an element of the form

$$(1, \tau)$$

with $\text{Im}(\tau) > 0$. Therefore

Theorem 7.7 *$\mathbf{T}(\Gamma, \text{Aut}(\mathbb{C}))$ is identified with the upper half plane.*

7.2.2 Moduli space

We need to determine $\text{Aut}(\Gamma) = \text{Aut}(\mathbb{Z} \oplus \mathbb{Z})$. Fixing generators γ_1 and γ_2 for Γ , new generators are determined clearly by a matrix in $SL(2, \mathbb{Z})$. To find the action $SL(2, \mathbb{Z}) \times \mathbf{T}(\Gamma) \rightarrow \mathbf{T}(\Gamma)$ we act

$$\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{Z}),$$

where a, b, c, d are integers with $ad - bc = 1$, on an element $(1, \tau)$ and obtain then

Theorem 7.8 *Two tori determined by τ and τ' are biholomorphic if and only if*

$$\tau' = \frac{a\tau + b}{c\tau + d}.$$

So obtain,

Corollary 7.9 $M(\Gamma, \text{Aut}(\mathbb{C})) = H/PSL(2, \mathbb{Z})$.